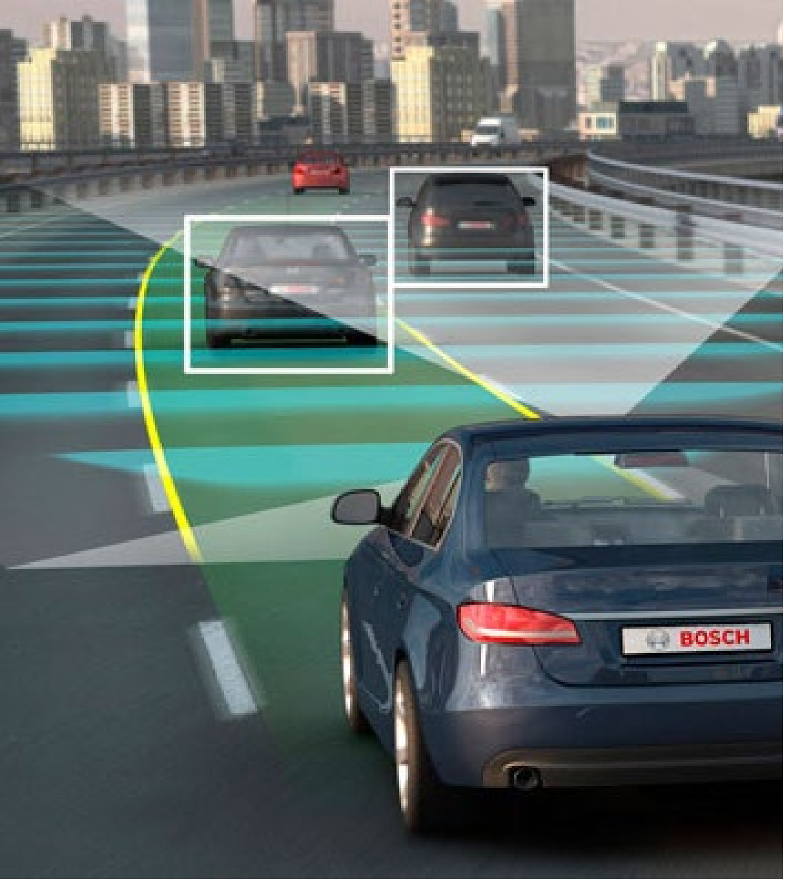

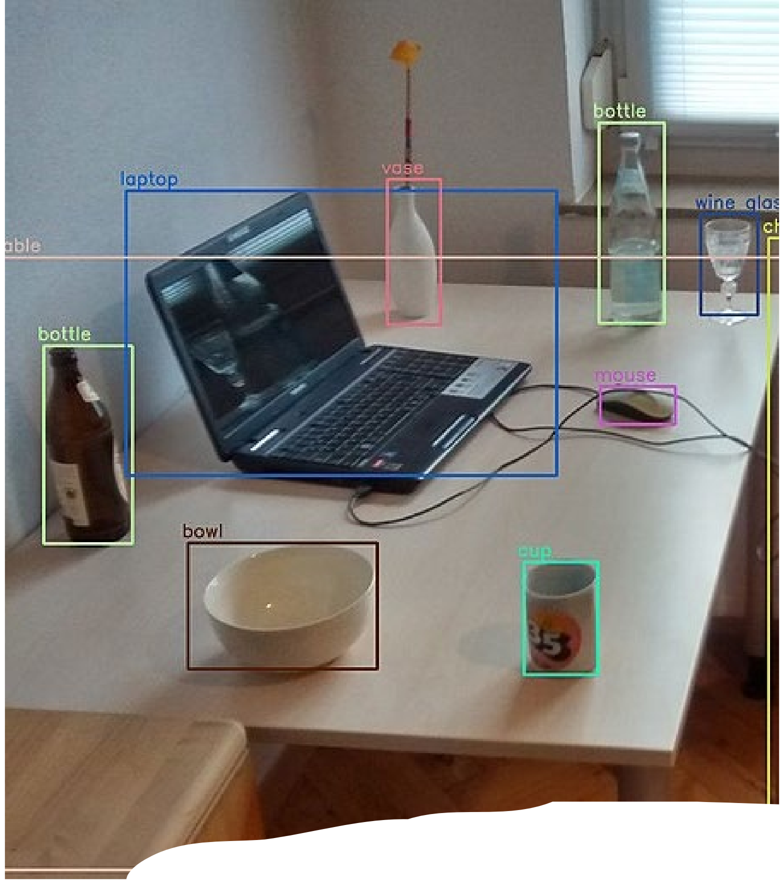

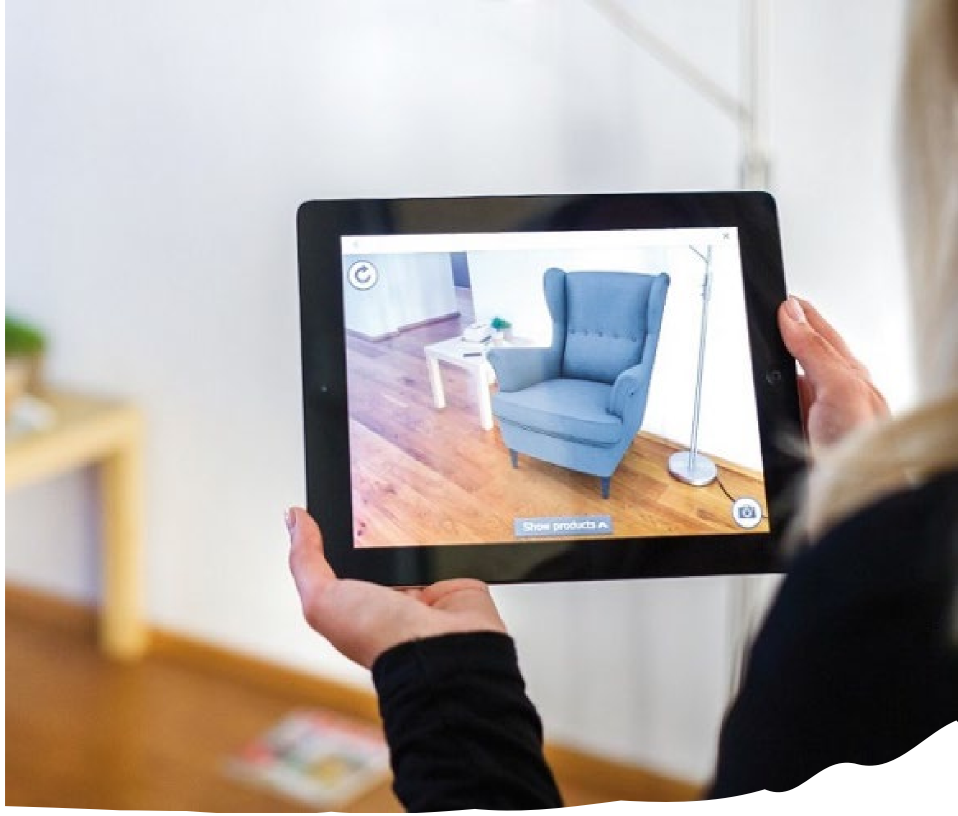

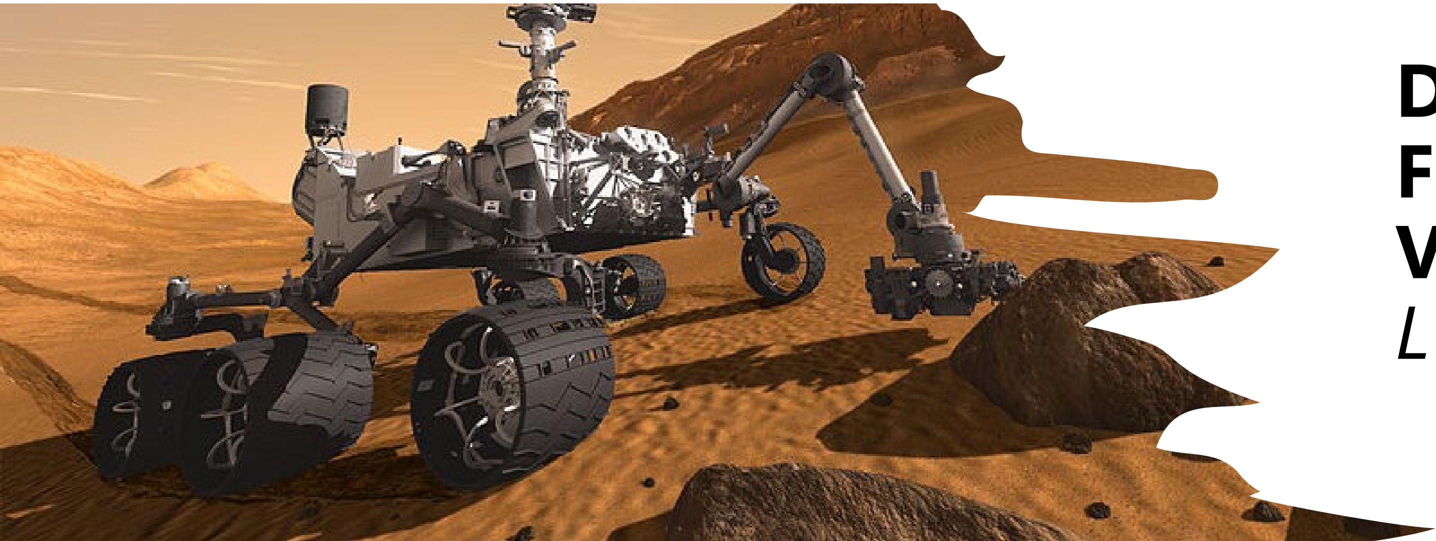

DT8055
Fundamentals of Computer Vision with Deep Learning

L1: Introduction

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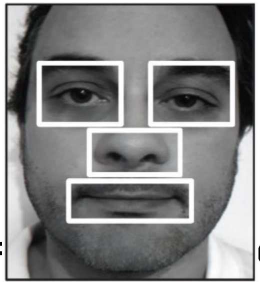
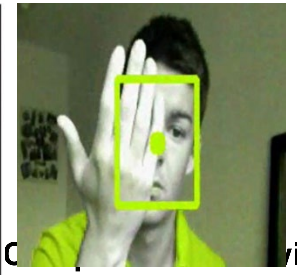
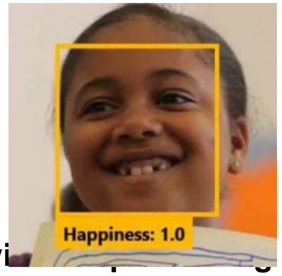
Teachers

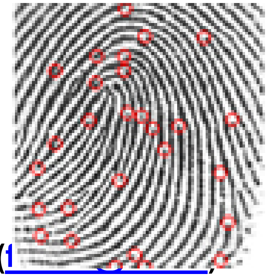
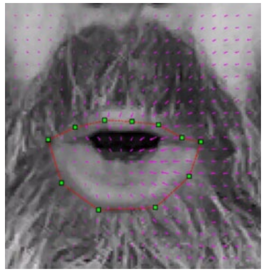
Fernando Alonso-Fernandez
(feralo@hh.se)

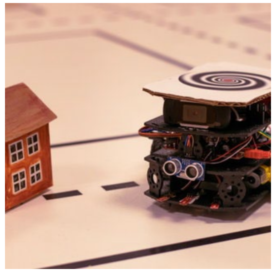
Kevin Hernandez-Diaz
(kevin.hernandez-diaz@hh.se)

Thu Ngo
(thu.ngo@hh.se)

- Research interests (recent, selection):
  - Biometrics (face, iris, ocular, fingerprints)
  - Soft-biometrics, emotion
  - Mobile biometrics, social robots
  - Autonomous driving

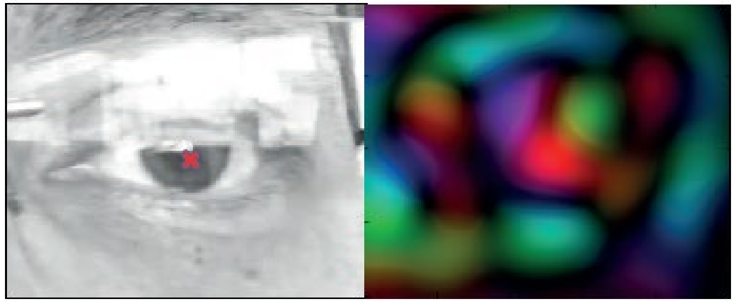





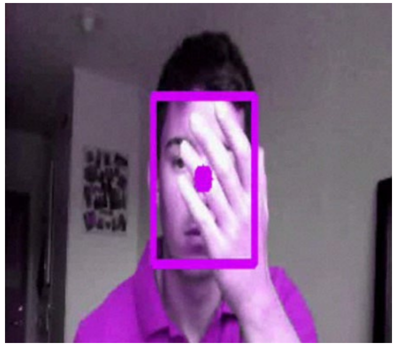

Some of our work at Halmstad

- <u>Biometrics:</u> detection & recognition

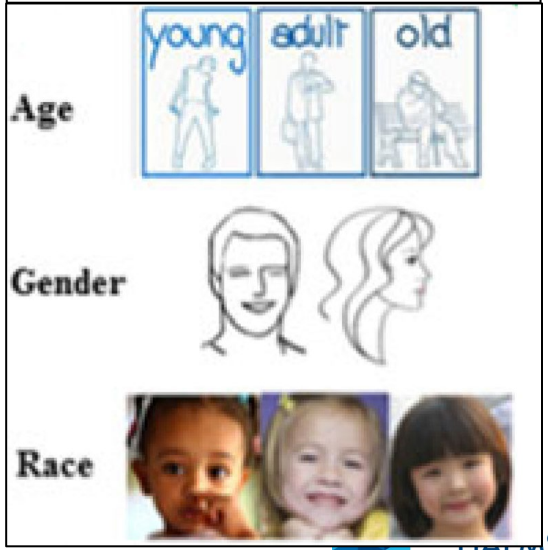
eye detection



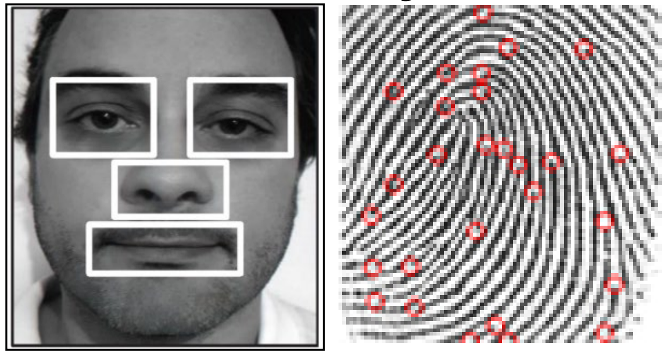
tracking



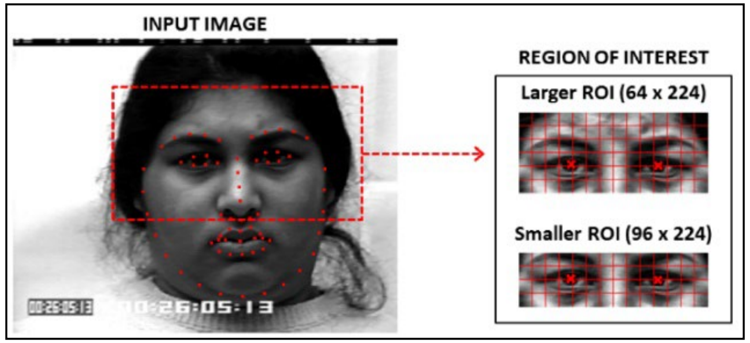
demographics



identity



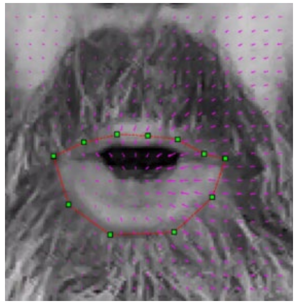
expression



Some of our work at Halmstad

- Biometrics: real-world issues

user interaction



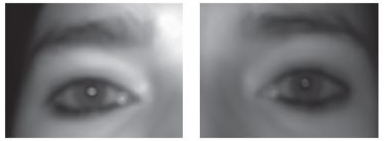

mobile operation


cross-spectral recognition

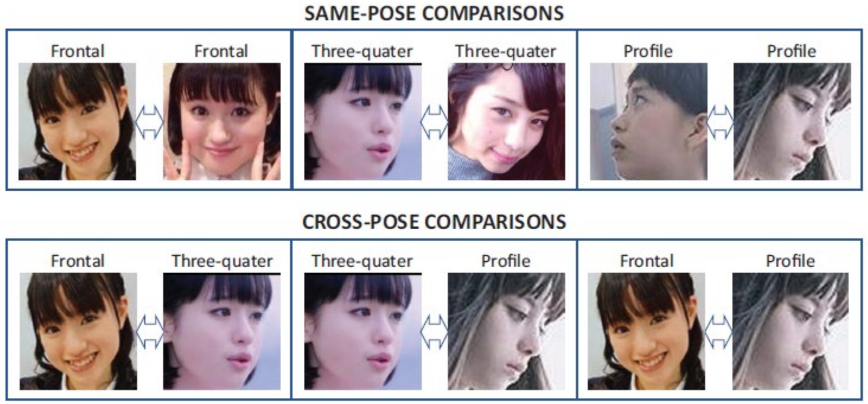
visible

night vision

infra-red



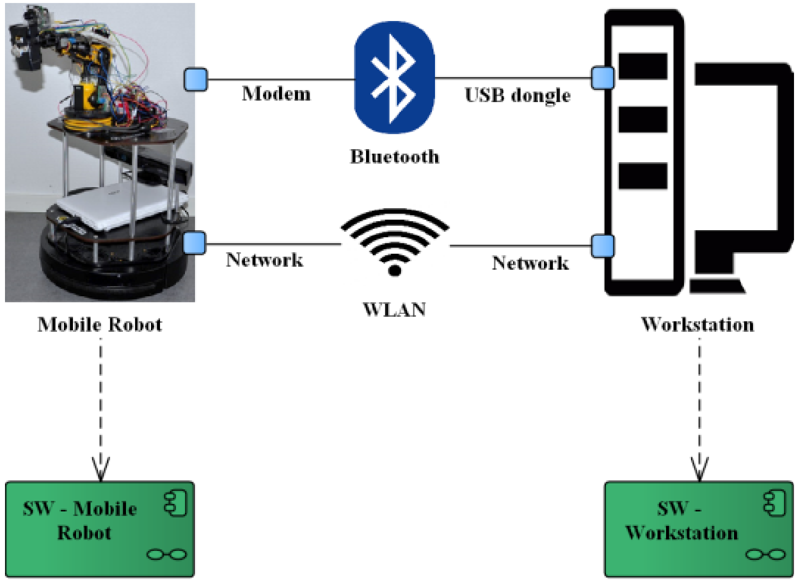
cross-pose recognition



Some of our work at Halmstad

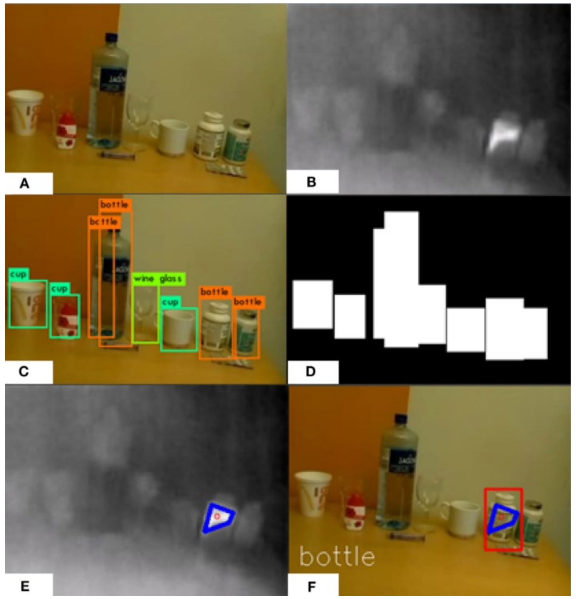
- Social robotics

first aid



exercise training

medicine intake



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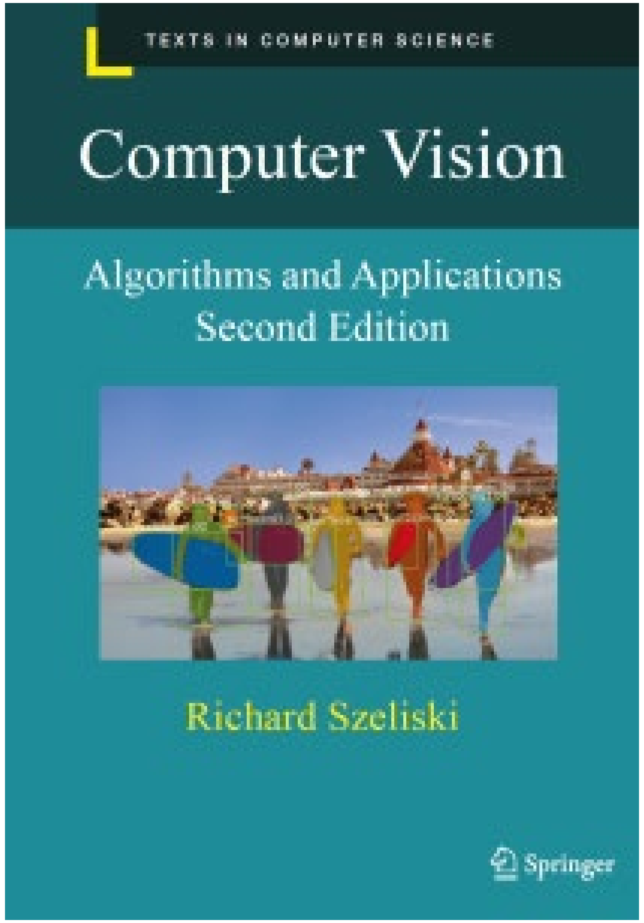
Some of our work at Halmstad

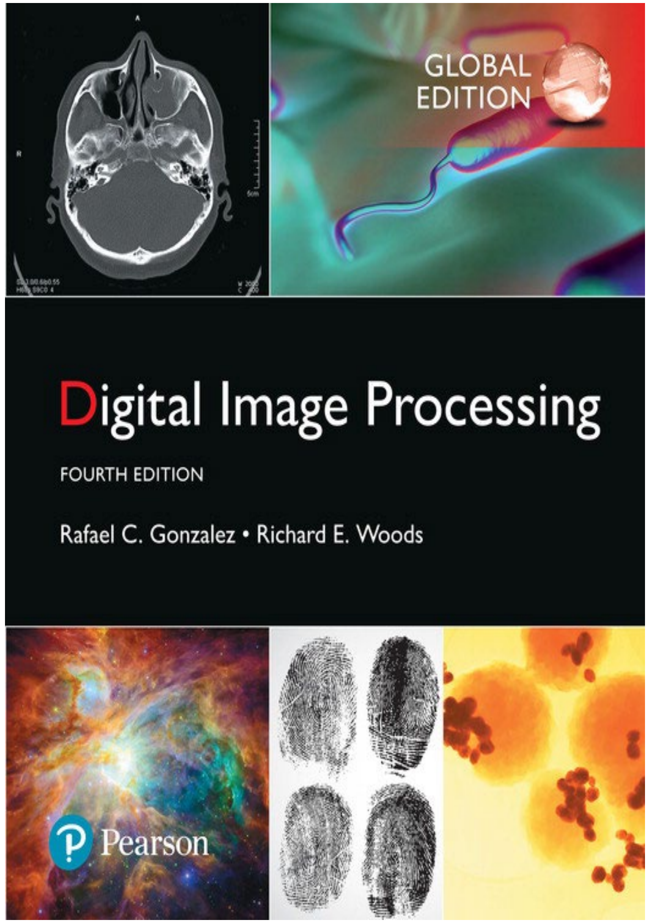
- Autonomous driving

semantic segmentation with LIDAR

Materials

- Rick Szeliski, *Computer Vision: Algorithms and Applications* (2022, 2nd ed) online at: http://szeliski.org/Book/
- Rafael C. Gonzalez, Richard E. Woods: *Digital Image Processing*. Global Edition, 4th Edition, Pearson 2018 (https://www.imageprocessingplace.com/)
- Blackboard page





Course requirements

- Prerequisites (desirable)
  - Programming (Python)
  - Data structures
  - Linear algebra
  - Vector calculus

- Course does ***not*** assume prior imaging experience
  - computer vision, image processing, graphics, etc.

Grading

Three assignments

- Practical Assignment I: Image Acquisition and Analysis (1 c)
- Practical Assignment II: Feature Extraction and Classification (2 c)
- Practical Assignment III: Computer Vision Applications (2 c)

NO EXAMS (Yay!)

Acknowledgements

Many thanks to:

- CS5670 (Intro to CV, Cornell Tech)
- CS231n (Deep Learning for Computer Vision, Stanford)
- CSE576 (Computer Vision, University of Washington)
- https://www.imageprocessingplace.com
- http://szeliski.org/Book

for contributing to having finished the materials of this course in a finite time, thus keeping sanity of the teacher under control

Today

1. Course overview (√)
2. What is computer vision? *vs. Machine Learning, Deep Learning, AI*
3. Tasks and applications of computer vision
4. Why study computer vision? *Because it's cool…*
5. How hard is computer vision? *Ehm… more than the current hype suggests*
6. Computer vision levels

Readings for today

- Gonzalez & Woods, Chapter 1 (Introduction)
- Szeliski, Chapter 1.1 (mandatory), 1.2 (optional)

Every image tells a story

- Goal of computer vision: perceive the "story" behind the **picture**, just like humans do

- Compute properties of the world
  - 3D shape, motion
  - Names and number of people or objects
  - What happened or will happen?
  - Action, intention, categories

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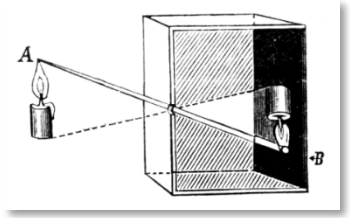
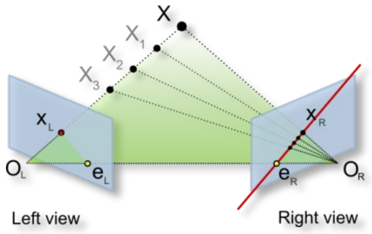
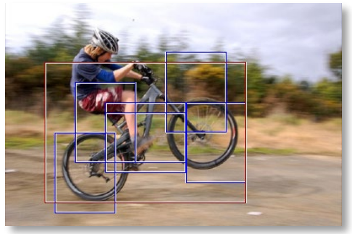
Computer Vision



1. Low-level vision: *primitive* operations
   – Image acquisition and representation, transformations, enhancements, convolution and filtering, edges and lines…

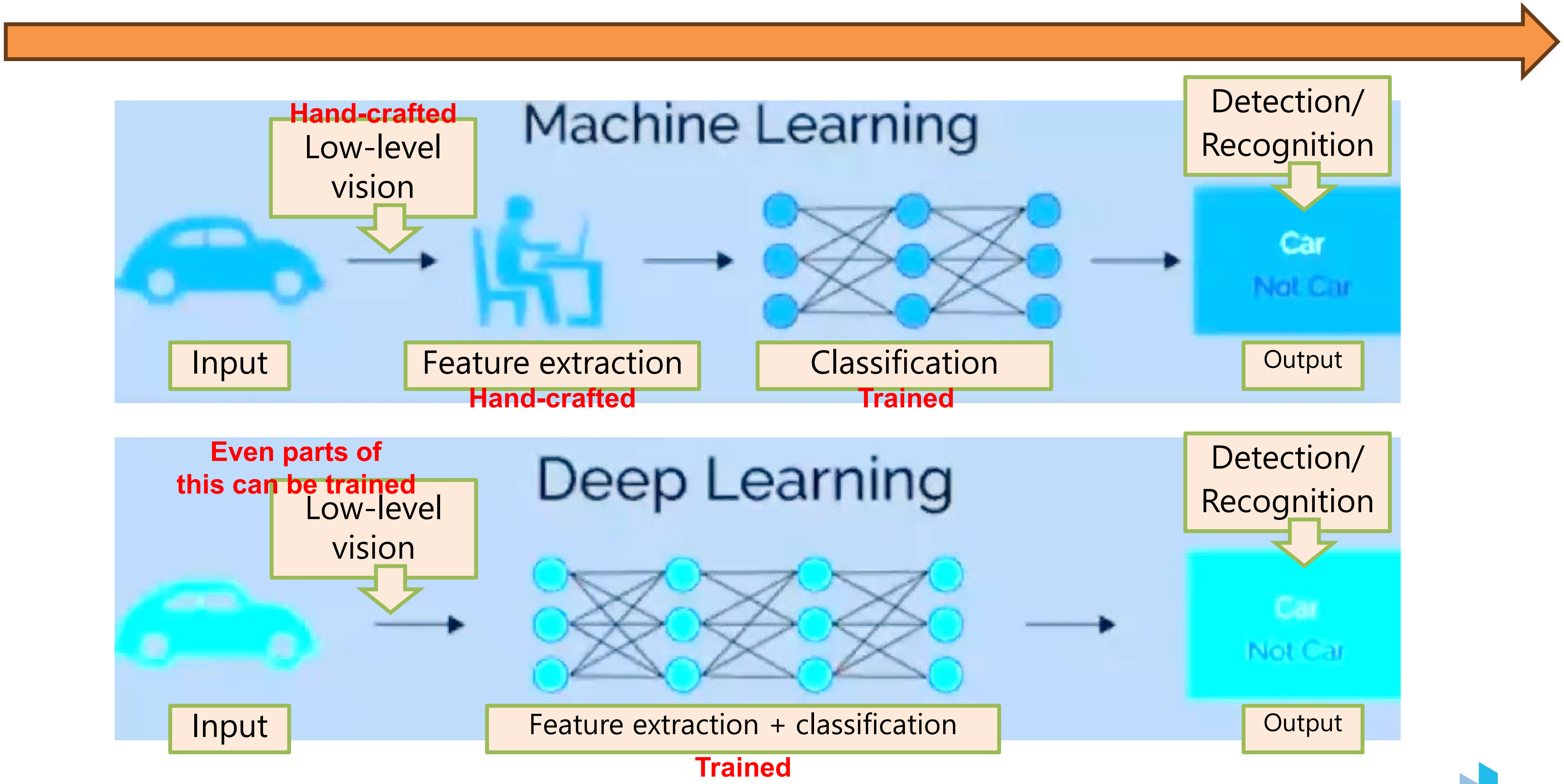


2. Mid-level vision: *attribute* extraction
   – Segmentation, depth, motion, 3D…
   – Feature extraction, classification, image matching…
   – Machine Learning (ML), Convolutional Neural Nets (CNN)



3. High-level vision: *making sense* of the image
   – Object / category / activity… detection and recognition

Machine Learning (ML) vs. Deep Learning (DL)



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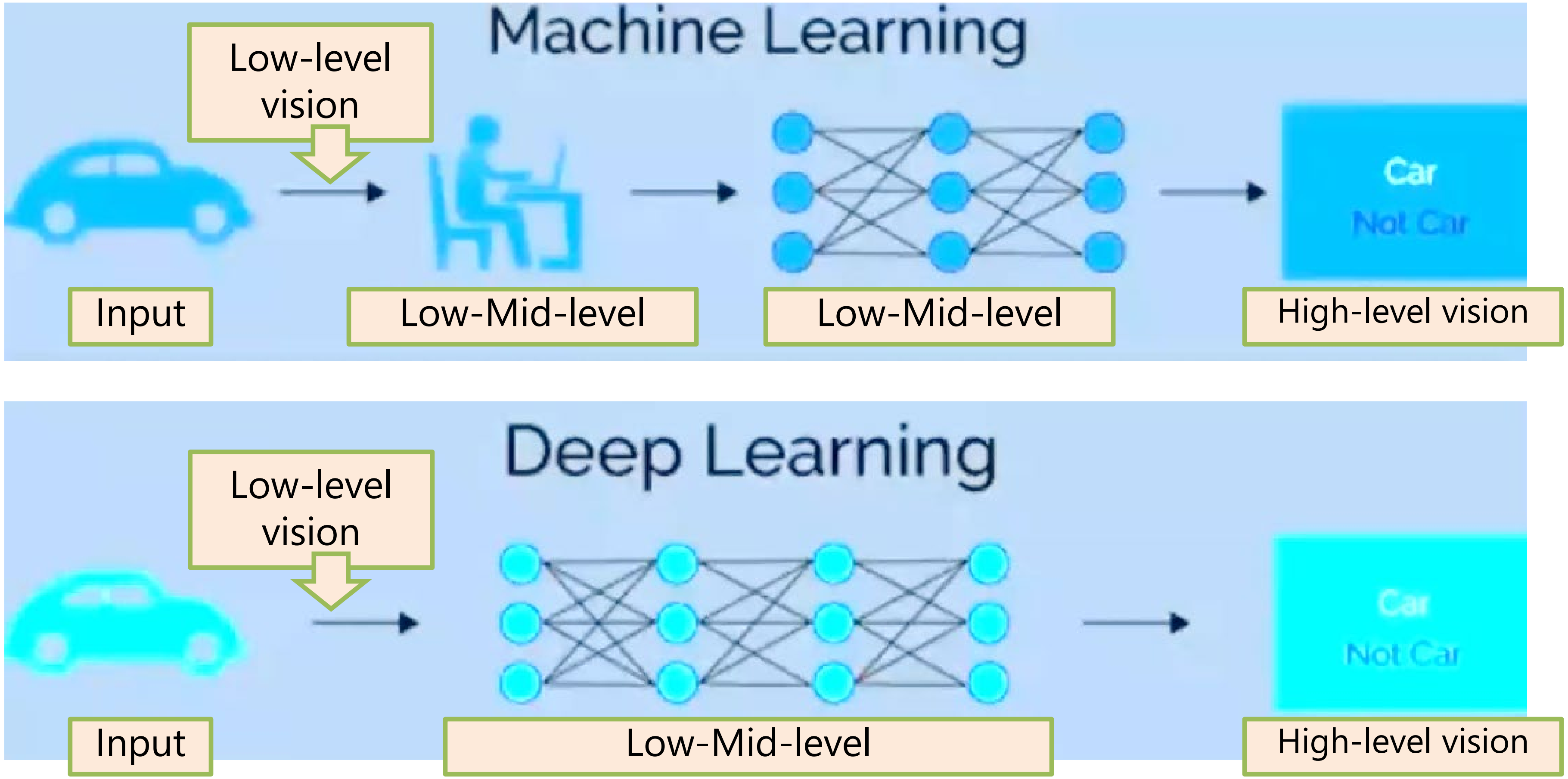
Machine Learning (ML) vs. Deep Learning (DL)



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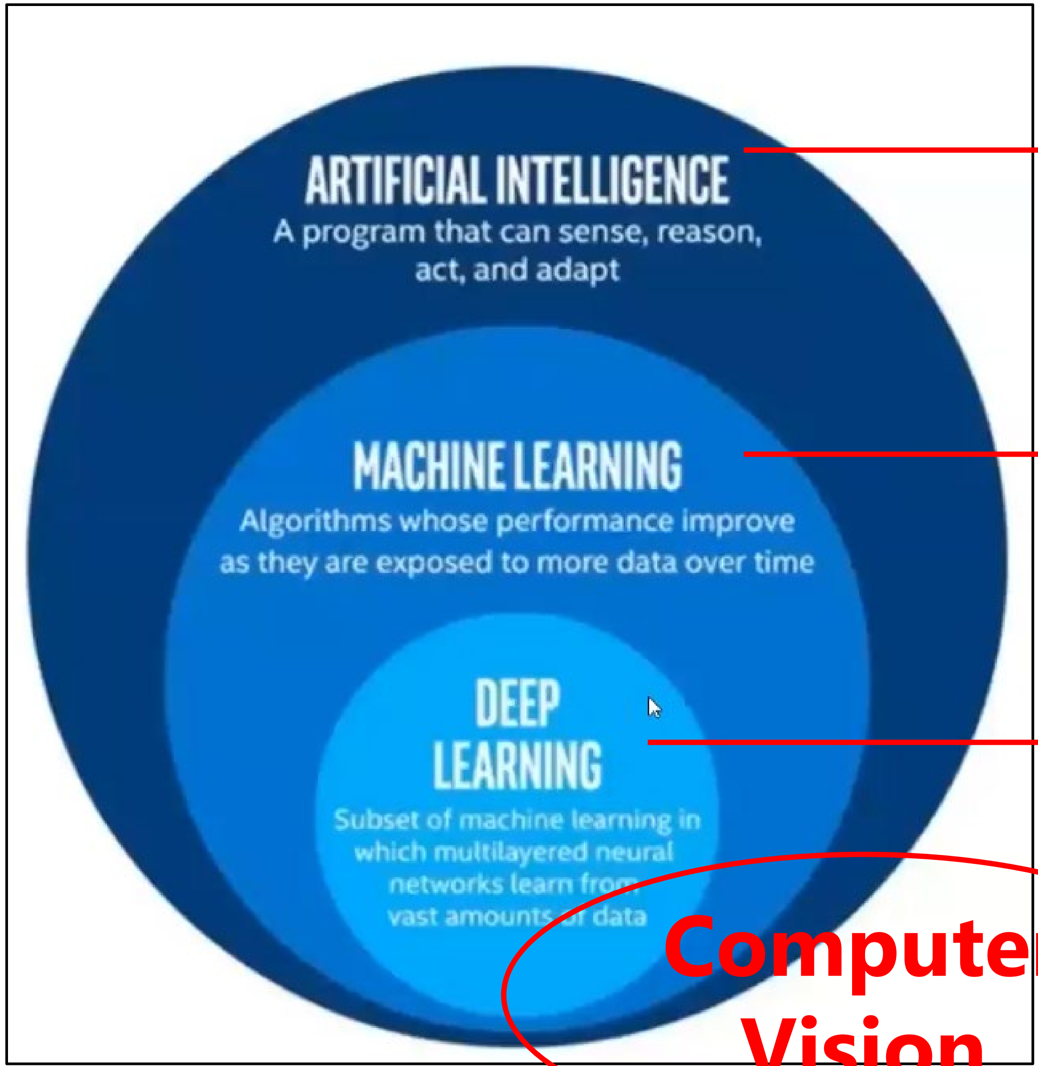
Artificial Intelligence (AI)



- Use computers to do tasks normally requiring *human intelligence* (visual perception, speech, reasoning, judgement, decision-making…)
- Does NOT mean doing a task in the same way as humans would do it, but doing it *smartly*

- Provide data to a computer to learn relationships and patterns in the data and be able to make predictions when new (*unseen*) data is presented
- In a (semi-) automatic, statistical way
- Improve over time with more data (i.e. *experience*)

- A particular way of training machine learning systems
- More complex (deep) models, tons of training data

Computer Vision

Making these tasks with images (there can be other types of data like 1D signals, audio, text…)

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Can computers match human perception?

- Yes and no (mainly no)
  - computers can be better at "easy" things
  - humans are better at "hard" things
- But huge progress
  - Accelerating in the last five years due to deep learning
  - What is considered "hard" keeps changing

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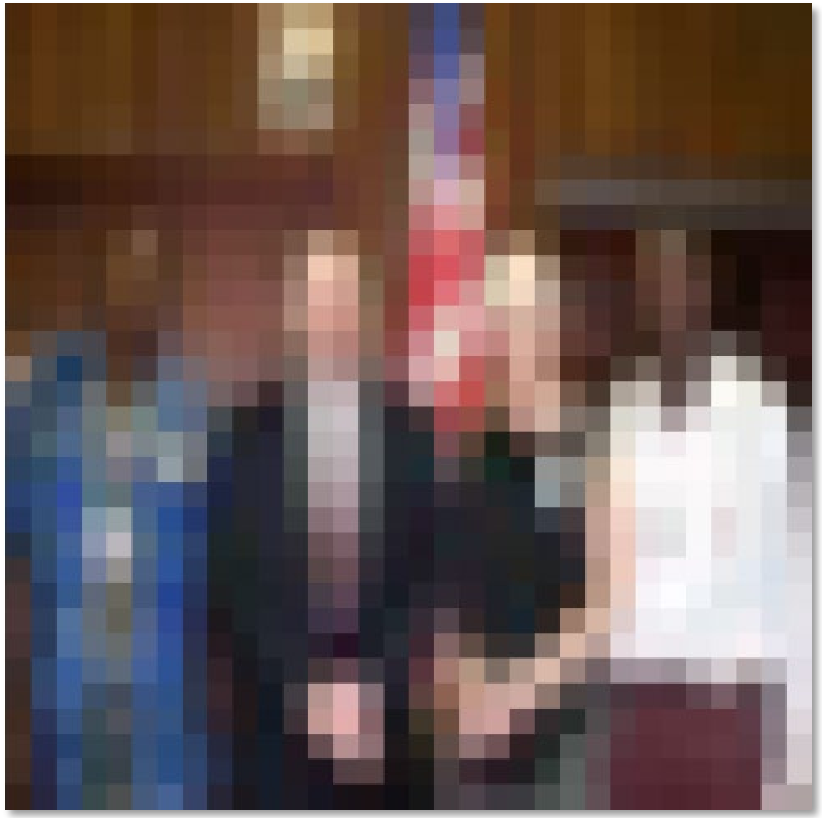
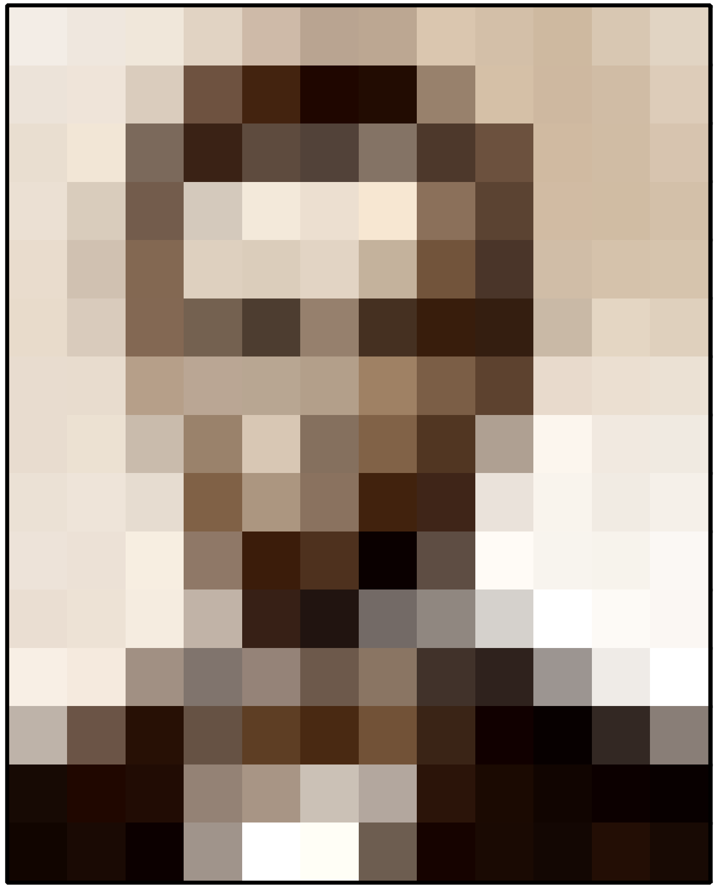
Humans can tell a lot about a scene from a little information…



Source: “80 million tiny images” by Torralba, et al.

What do computers see?



slide by Larry Zitnick

Tasks of computer vision (examples)

- Detection
- Segmentation
- Recognition
- 3D shape
- Depth
- Motion
- Image alignment
- Computational photography

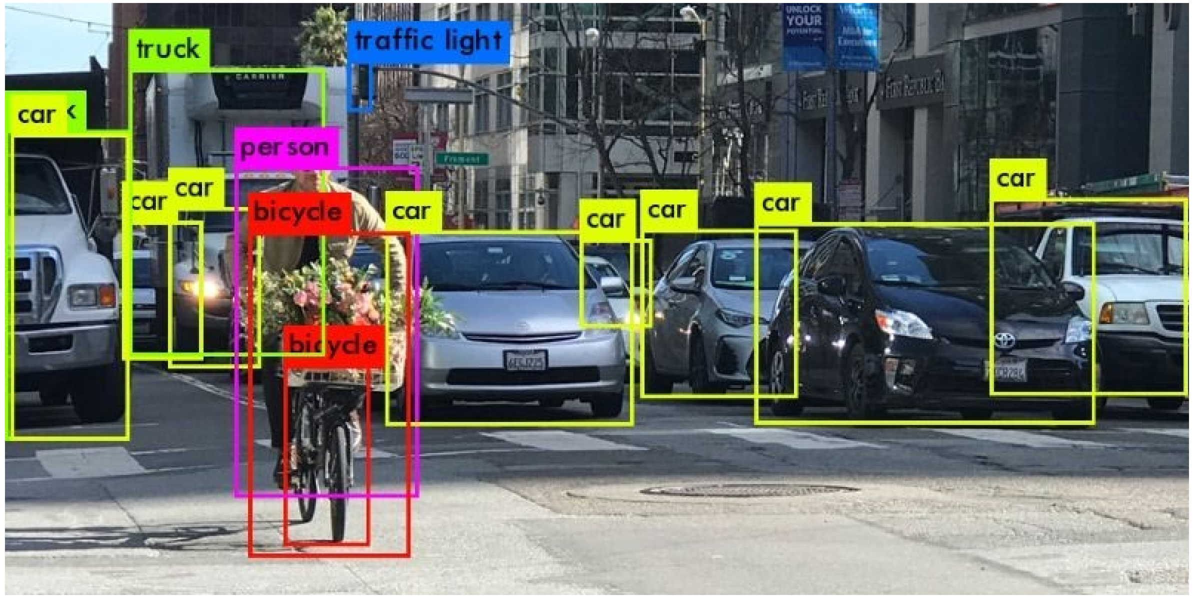
Tasks of computer vision

- Object detection

  (recognize objects generically, giving the smallest *bounding box* containing them)

- Semantic segmentation

  (find exact pixels occupied by objects)

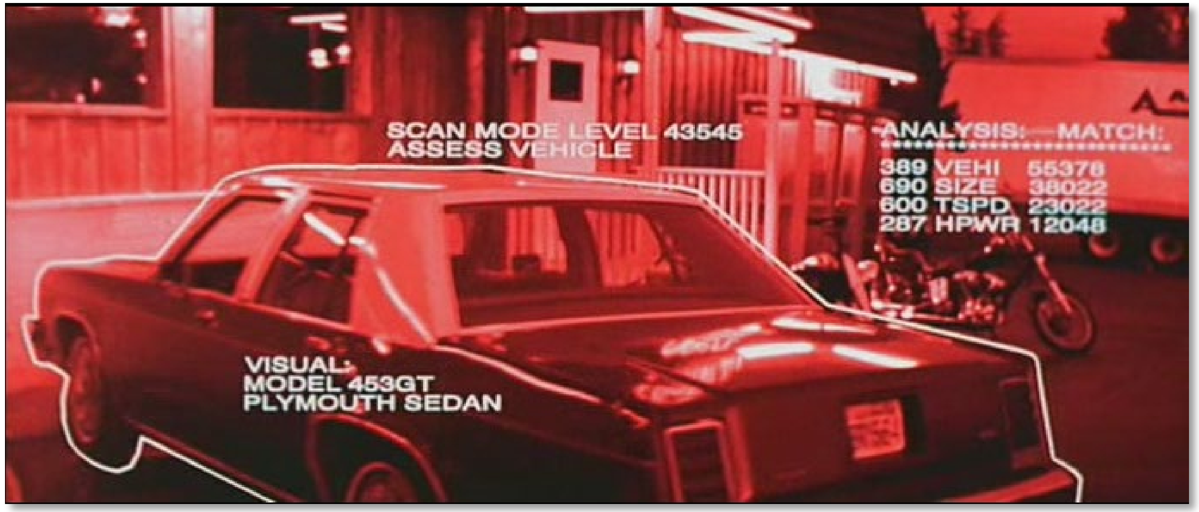


https://heartbeat.fritz.ai/introduction-to-basic-object-detection-algorithms-b77295a95a63

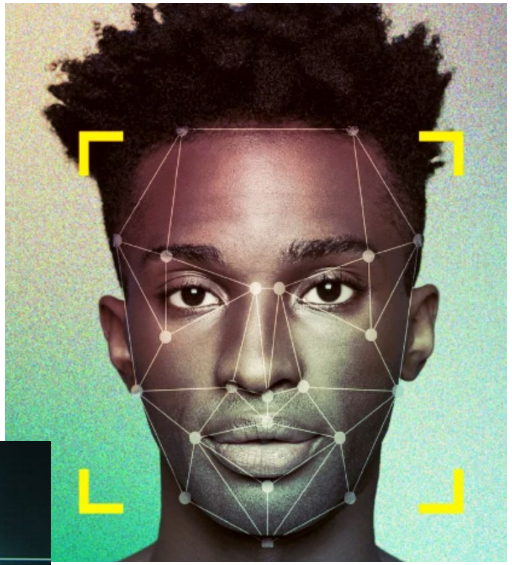
https://gts.ai/how-do-we-solve-the-challenges-faced-due-to-semantic-segmentation/

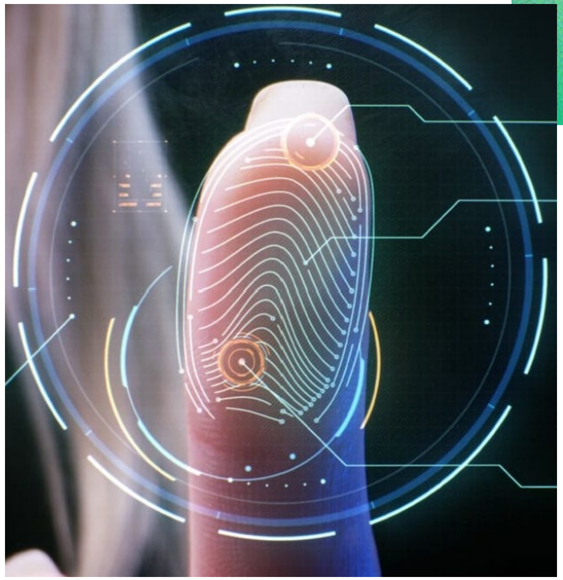
Tasks of computer vision

- Recognize (identify) specific objects or people



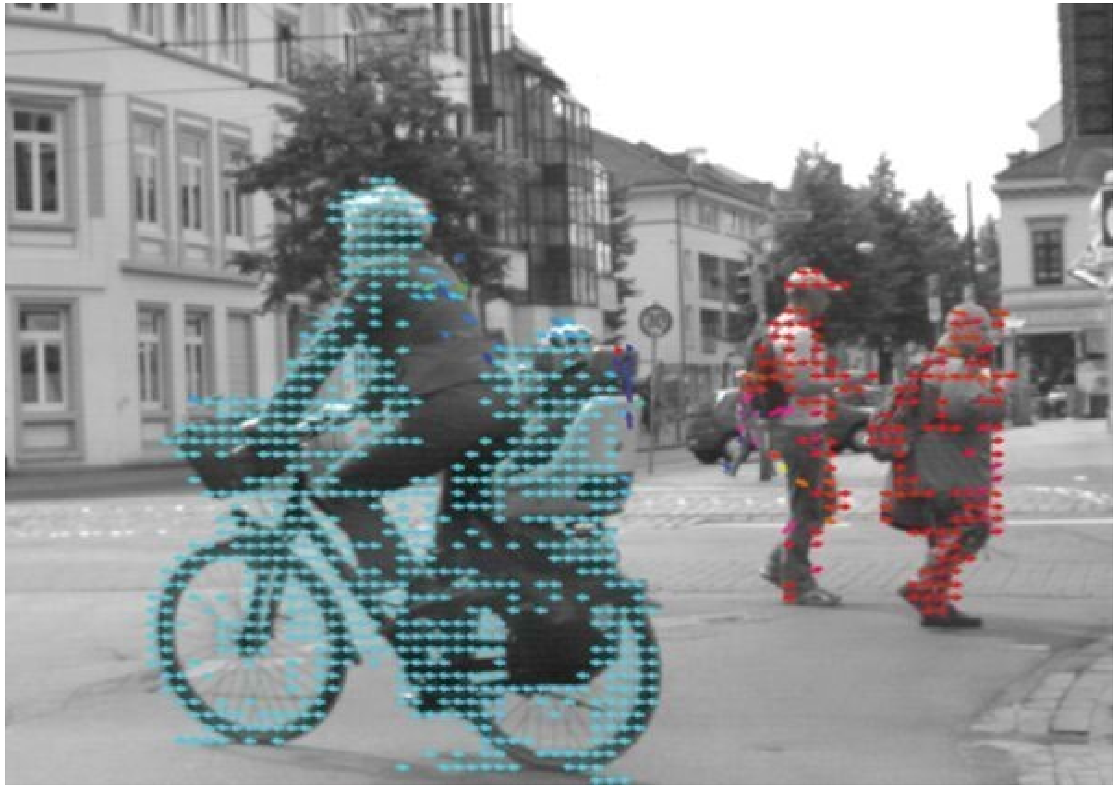
Terminator 2, 1991





Tasks of computer vision

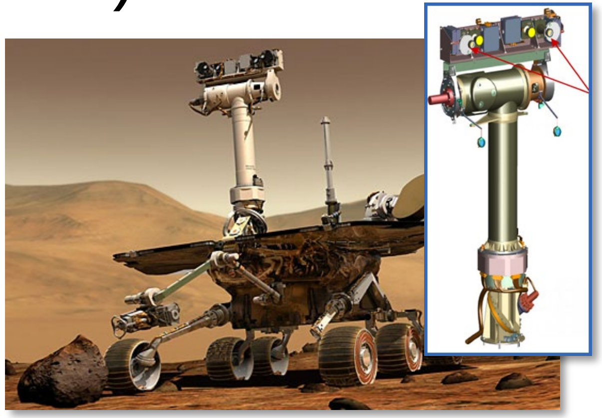
- Compute motion

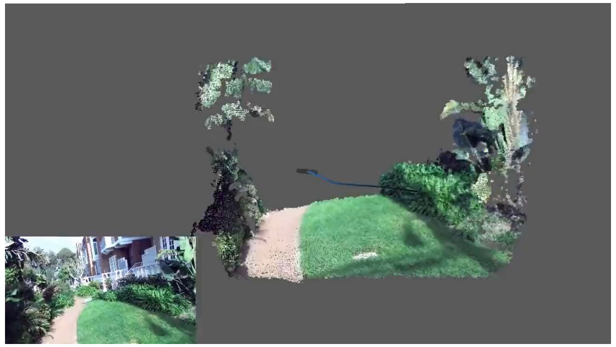


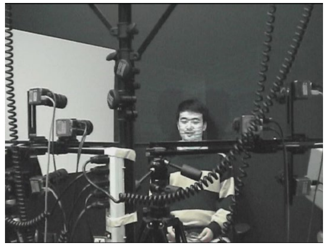
https://www.commonlounge.com/discussion/1c2eaa85265f47a3a0a8ff1ac5fbce51

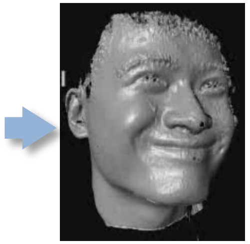
- Compute the 3D shape (and depth) of the world

ZED 2i Camera









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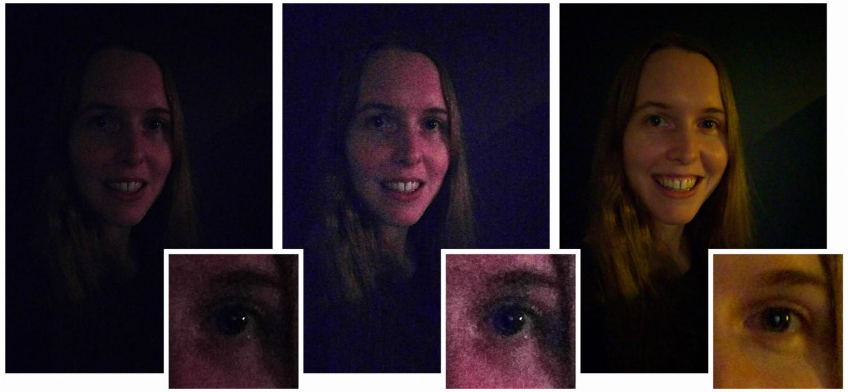
Tasks of computer vision

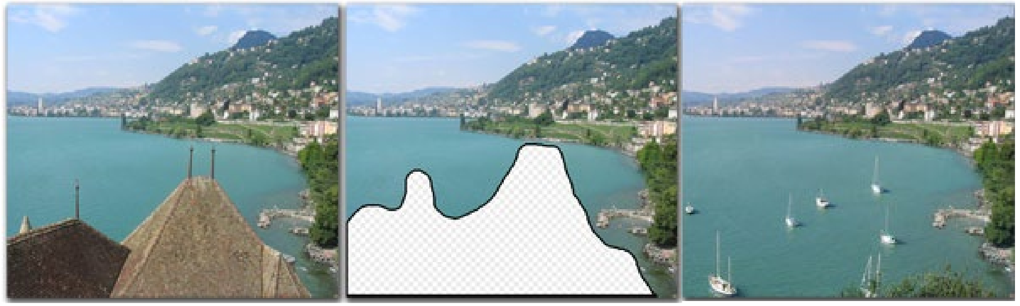
- Improve photos (“Computational Photography”)

Super-resolution (source: 2d3)



Low-light photography
(credit: Hasinoff et al., SIGGRAPH ASIA 2016)

Depth of field on cell phone camera
(source: Google Research Blog)



Inpainting / image completion
(image credit: Hays and Efros)

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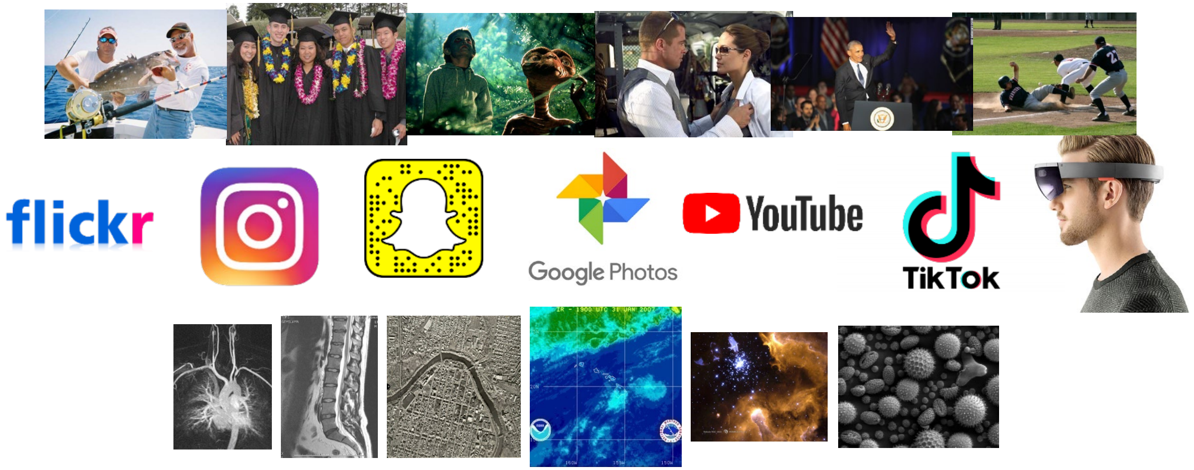
Why study computer vision?

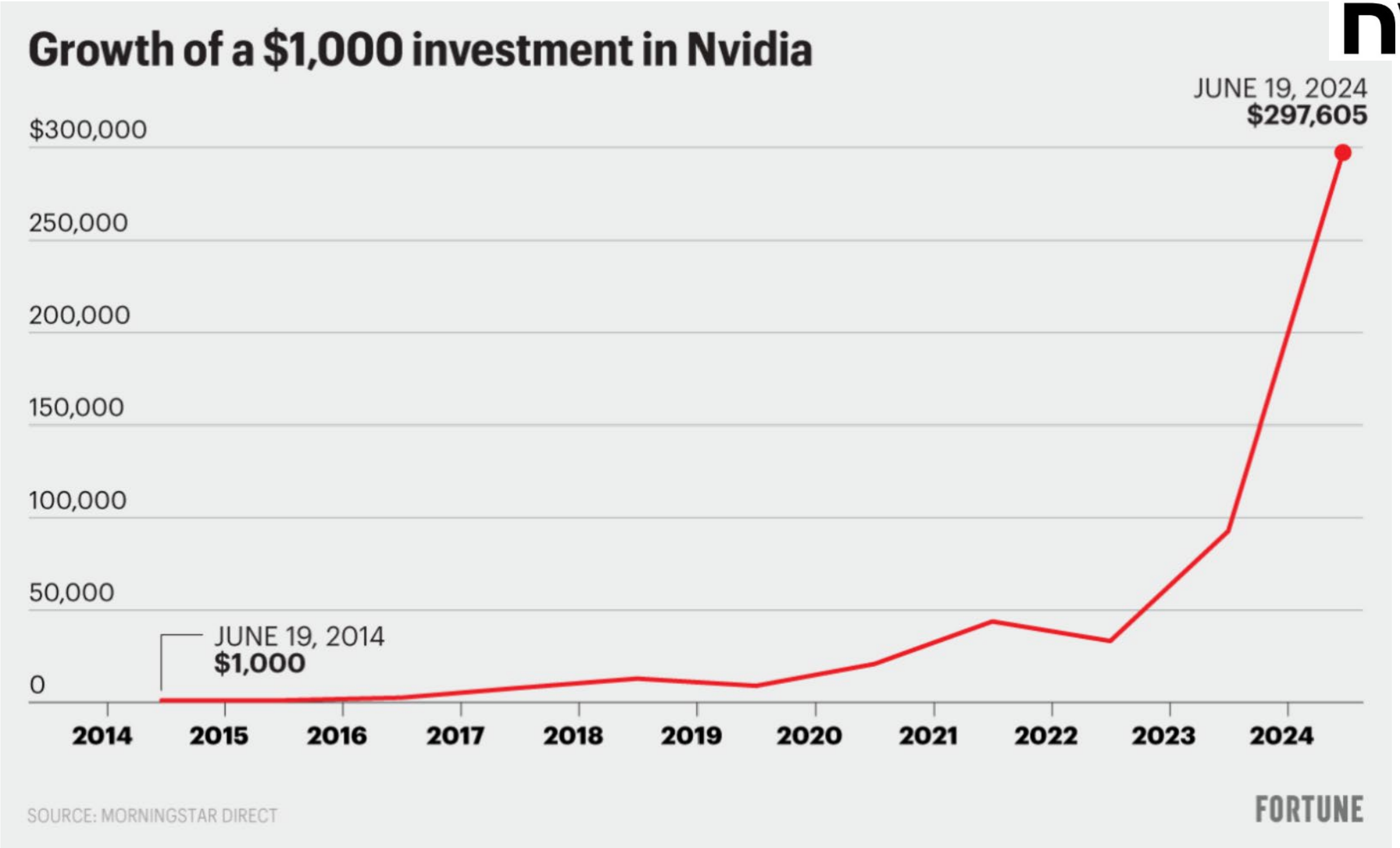
- Billions of images/videos captured per day
- Availability of powerful GPUs and cheap storage



- Huge number of potential applications
- The next slides show the current state of the art

Why study computer vision?

- Hype?



NVIDIA

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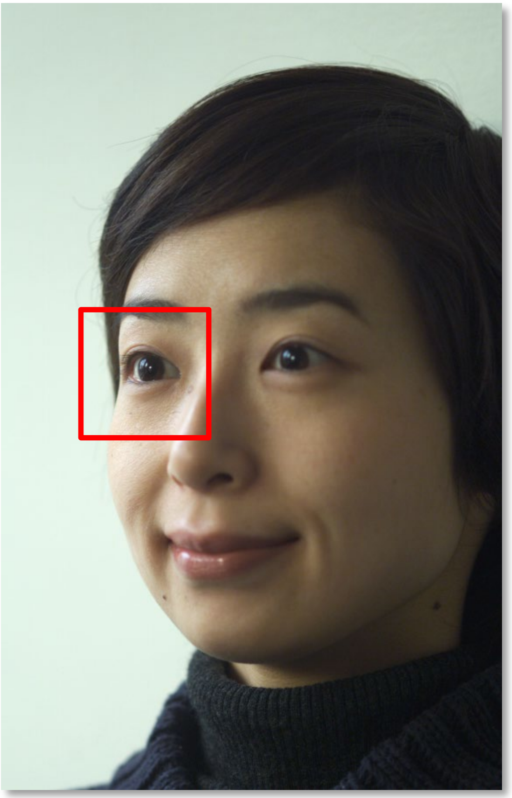
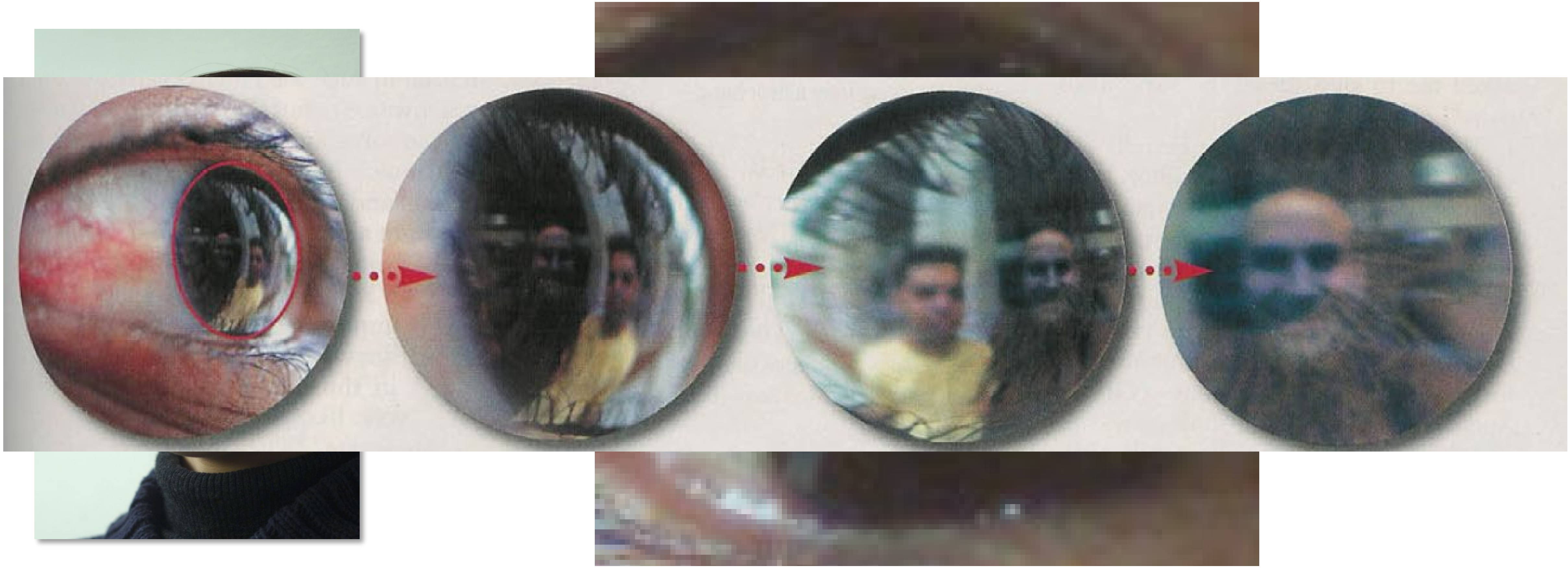
Applications of computer vision (examples)

Forensics



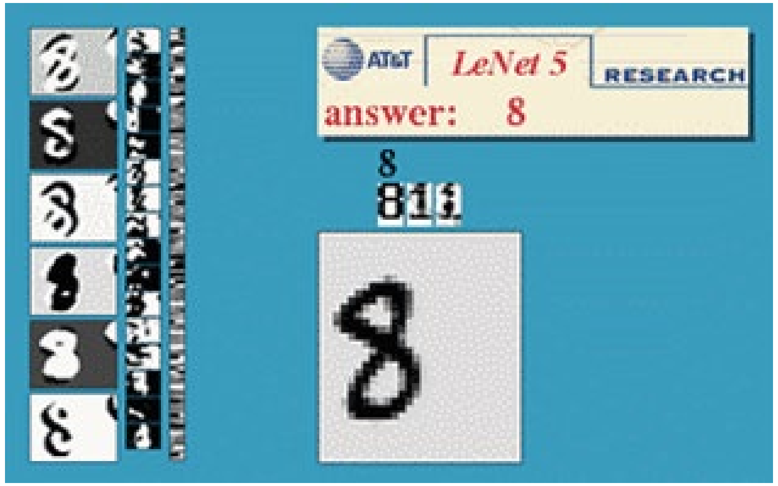
Source: Nayar and Nishino, “Eyes for Relighting”

Forensics



Source: Nayar and Nishino, “Eyes for Relighting”

Optical character recognition (OCR)



Digit recognition, AT&T labs (1990's)
http://yann.lecun.com/exdb/lenet/

License plate readers

Automatic text recognition

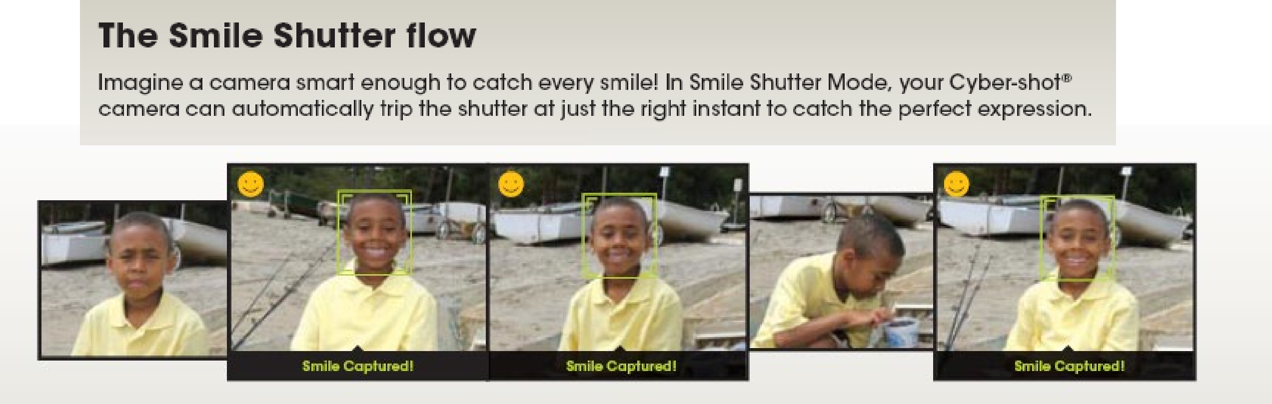
Automatic check processing

- Your scanner or telephone comes with OCR software

Face analysis

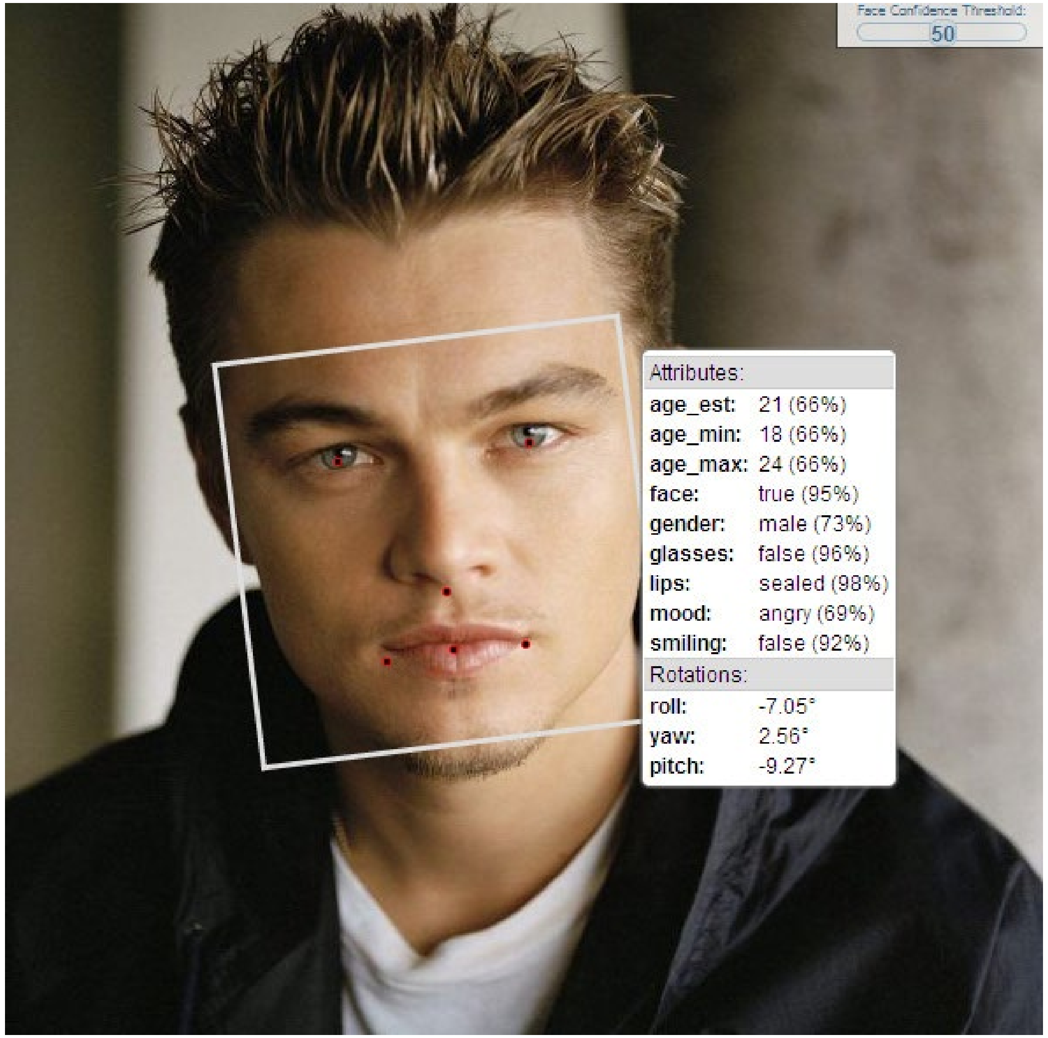
- Nearly all cameras detect faces in real time

Smile detection

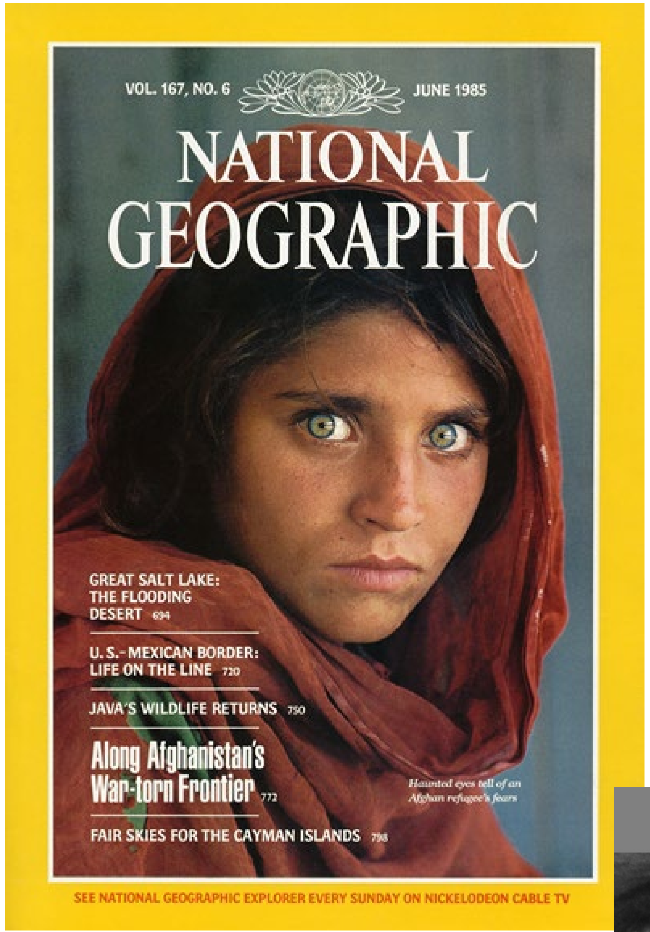


[Sony Cyber-shot® T70 Digital Still Camera](#) **Source: S. Seitz**

Demographics/expression

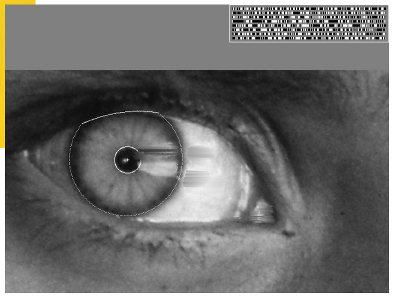


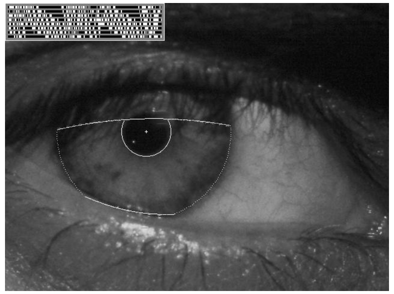
Biometrics

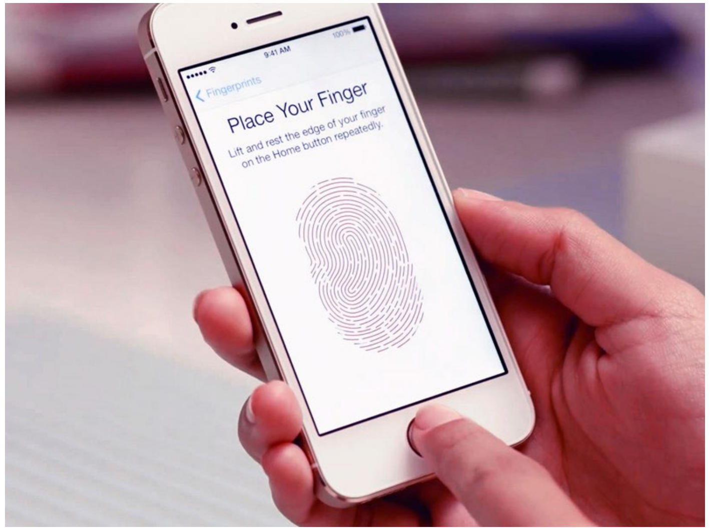


Who is she?

"How the Afghan Girl was Identified by Her Iris Patterns" Read the [story]







Fingerprint scanners on many new smartphones and other devices

Face unlock on Apple iPhone X
See also http://www.sensiblevision.com/

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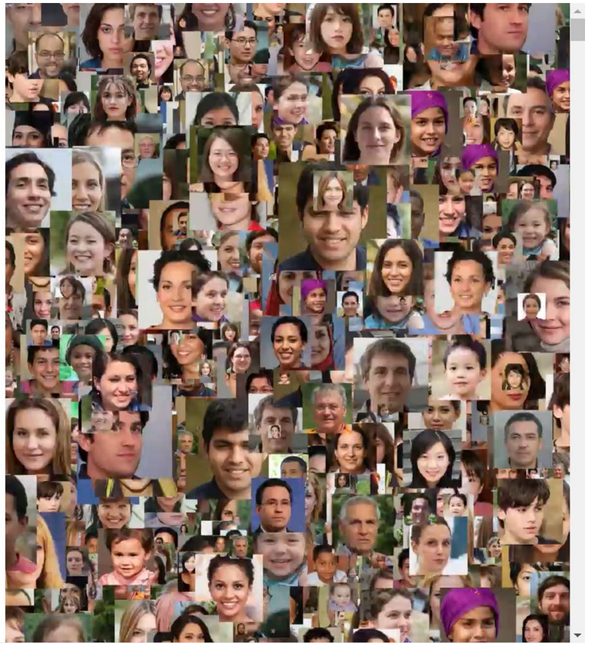
Privacy issues

The New York Times

Account

The Secretive Company That Might End Privacy as We Know It

A little-known start-up helps law enforcement match photos of unknown people to their online images — and "might lead to a dystopian future or something," a backer says.



New York Times, Jan. 18, 2020

by Kashmir Hill

Clearview.ai

Researchers warn peace sign photos could expose fingerprints

But the likelihood of anyone actually using images to recreate prints is pretty slim.

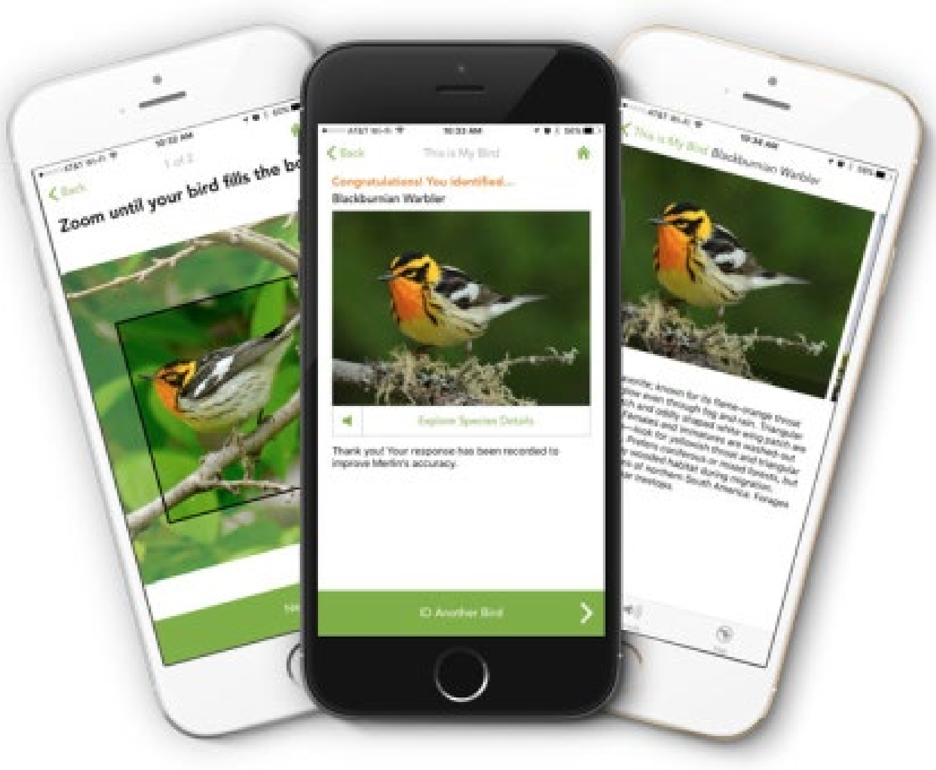
Jamie Rigg, @jmerigg
01.13.17 in Security

Comments

1721
Shares

Getty

Bird/plant identification



Merlin Bird ID (based on Cornell Tech technology)

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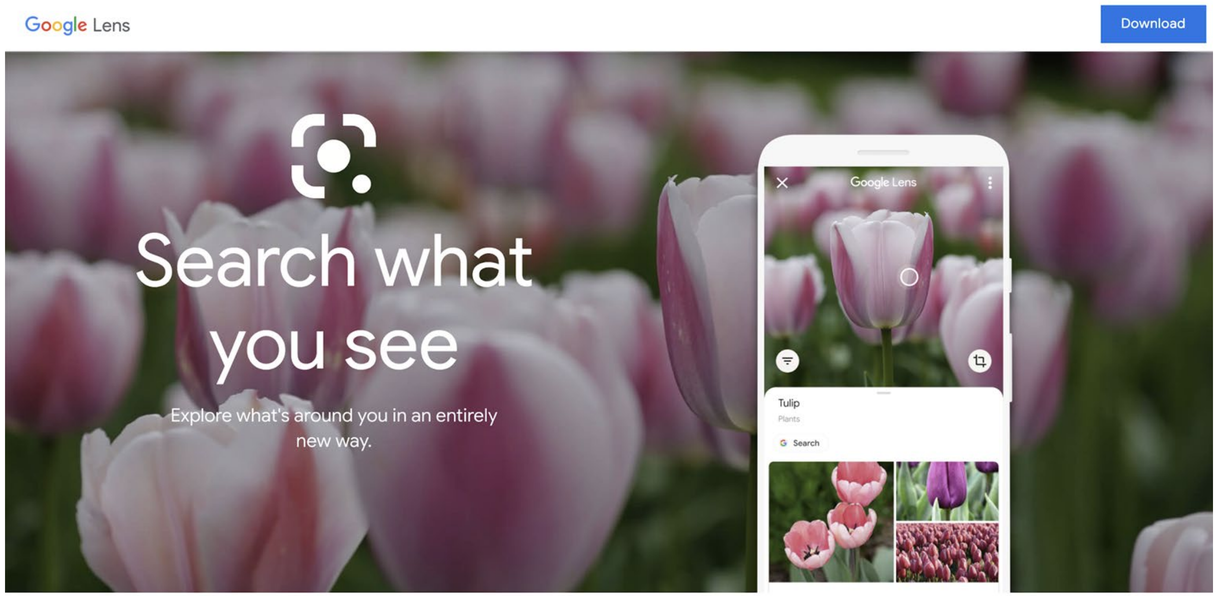
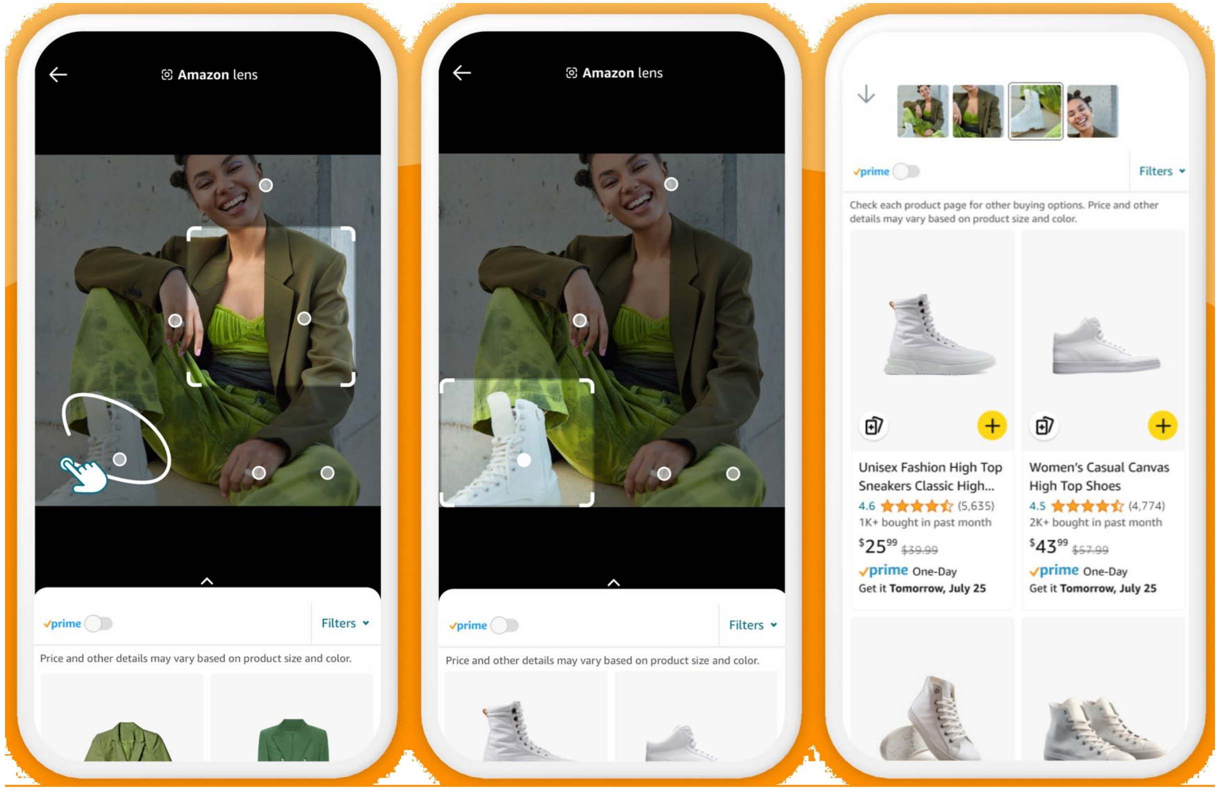
Object recognition





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Special effects: shape/motion capture

The Matrix movies, ESC Entertainment, XYZRGB, NRC

Pirates of the Carribean, Industrial Light and Magic

Source: S. Seitz

Los Angeles Times

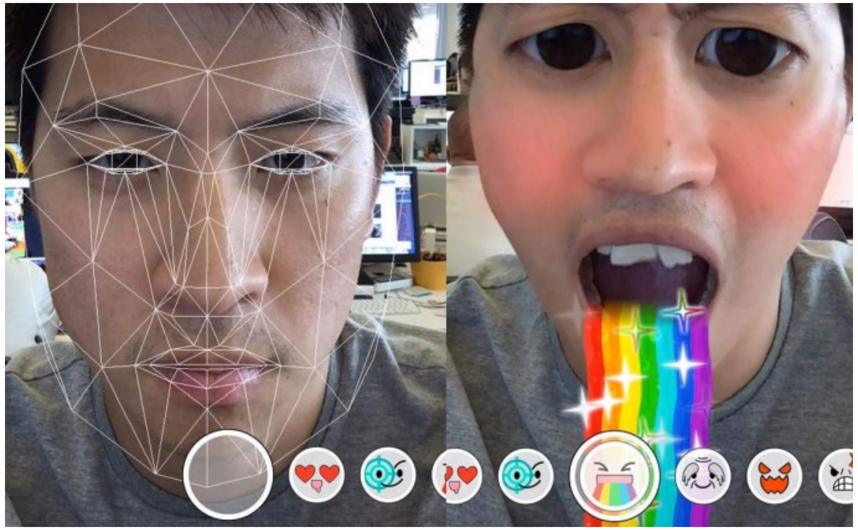
MOVIES

Robert De Niro said no green screen. No face dots. How ‘The Irishman’s’ de-aging changes Hollywood

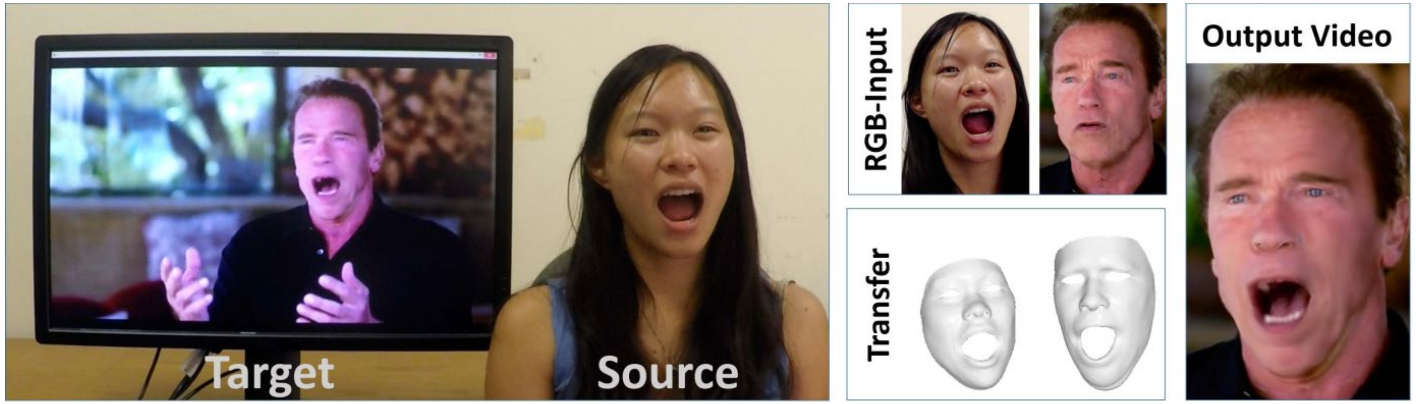
Makeup and wig work got Robert De Niro partway to his character, Frank Sheeran, at 41, left. It took a specially built camera and visual artists to get all the way there, as before-and-after images show. (Netflix)

By JOSH ROTTENBERG | STAFF WRITER JAN. 2, 2020

3D face tracking w/ consumer cameras



Snapchat Lenses

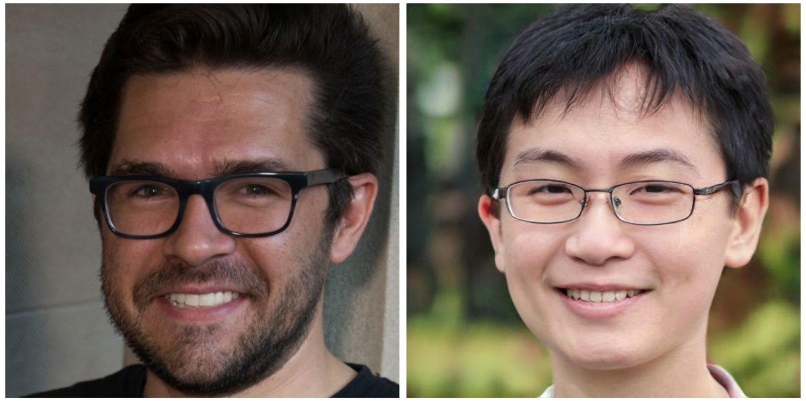


Face2Face system (Thies et al.)

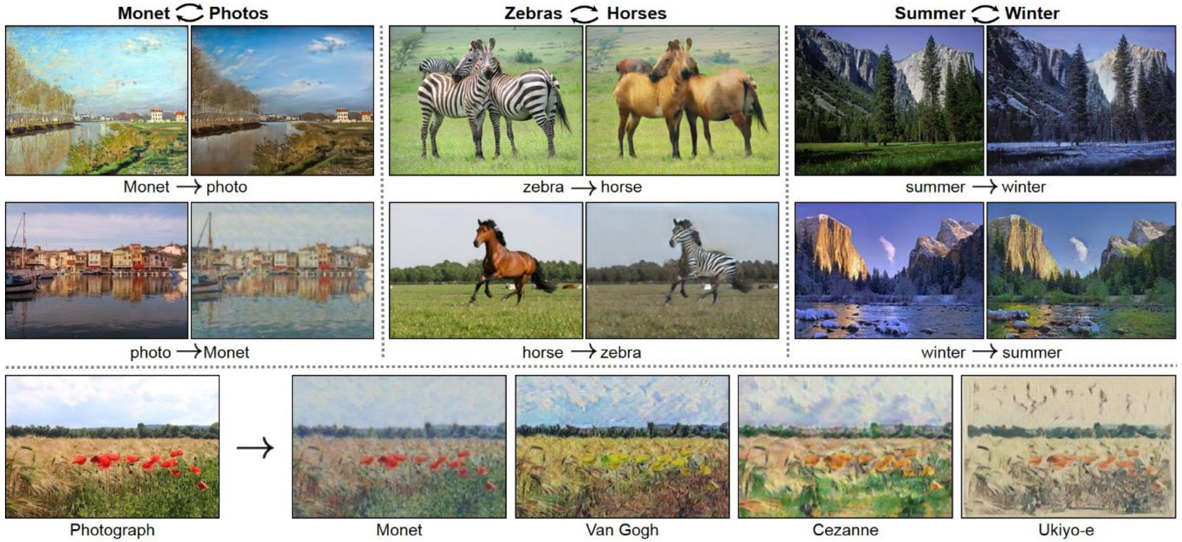
Image synthesis (generative AI)

Karras, et al., *Progressive Growing of GANs for Improved Quality, Stability, and Variation*, ICLR 2018

https://youtu.be/XOxxPcy5Gr4?si=19Mp8qLT10Kdn0Ot



https://www.whichfaceisreal.com/



Zhu, et al., *Unpaired Image-to-Image Translation using Cycle-Consistent Adversarial Networks*, ICCV 2017

Image synthesis (generative AI)

You

draw a nice image to show in the introductory lecture of a computer vision course

Copilot

I'll create a visual for you; one moment please.

Sports

Sportvision first down line
Explanation on www.howstuffworks.com

Source: S. Seitz

Smart cars / self-driving cars

- Mobileye
- Tesla Autopilot
- Safety features in many cars

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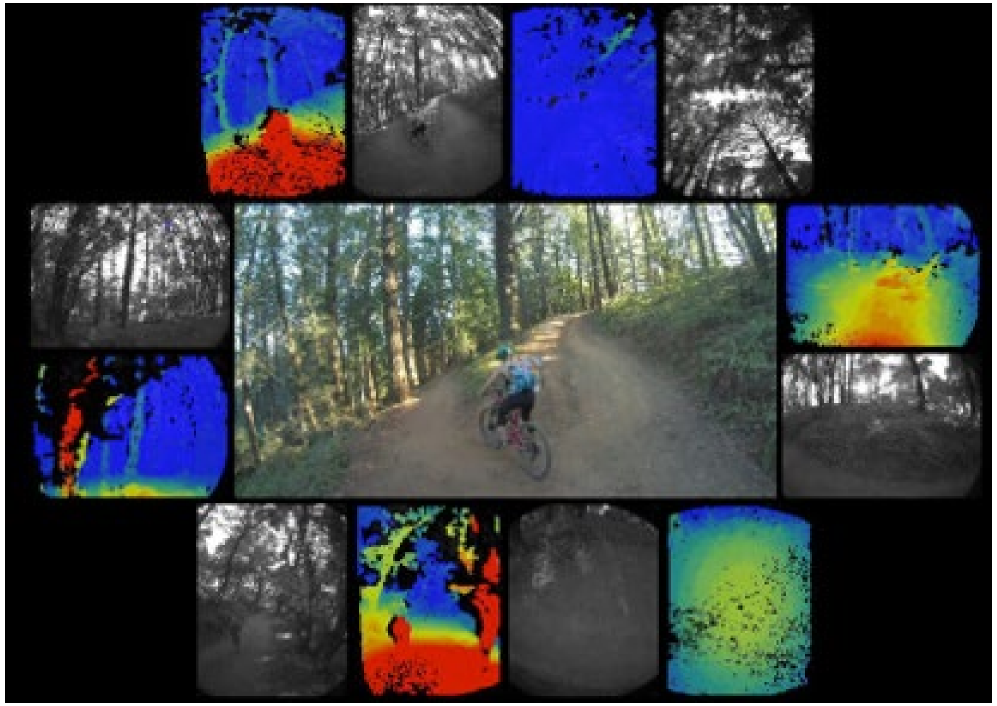
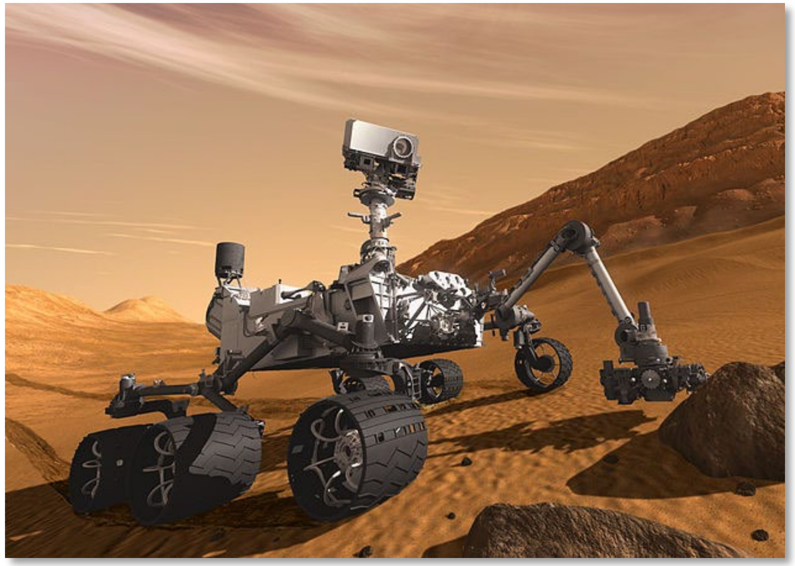
Drones



https://www.skydio.com/

Robotics



NASA's Mars Curiosity Rover
https://en.wikipedia.org/wiki/Curiosity_(rover)

Amazon Picking Challenge
http://www.robocup2016.org/en/events/amazon-picking-challenge/

Amazon Prime Air

Amazon Scout

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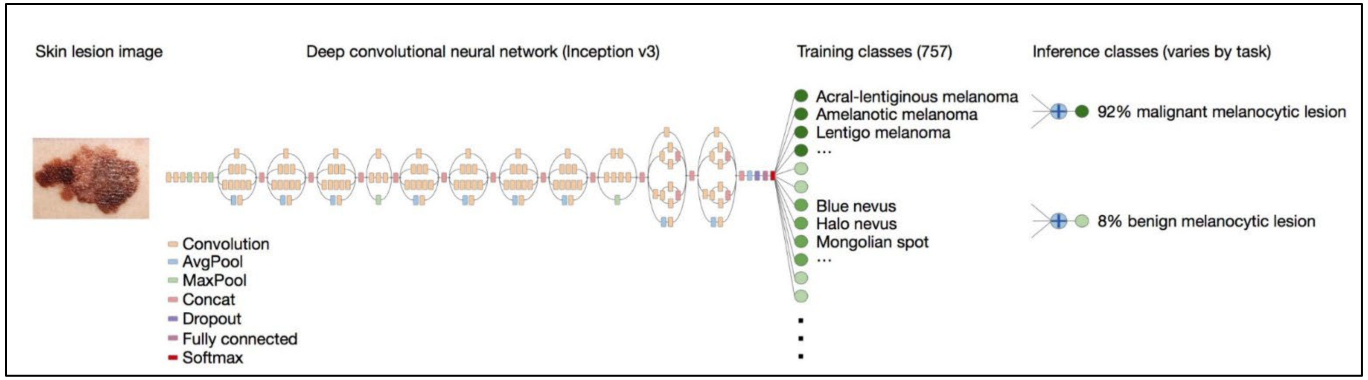
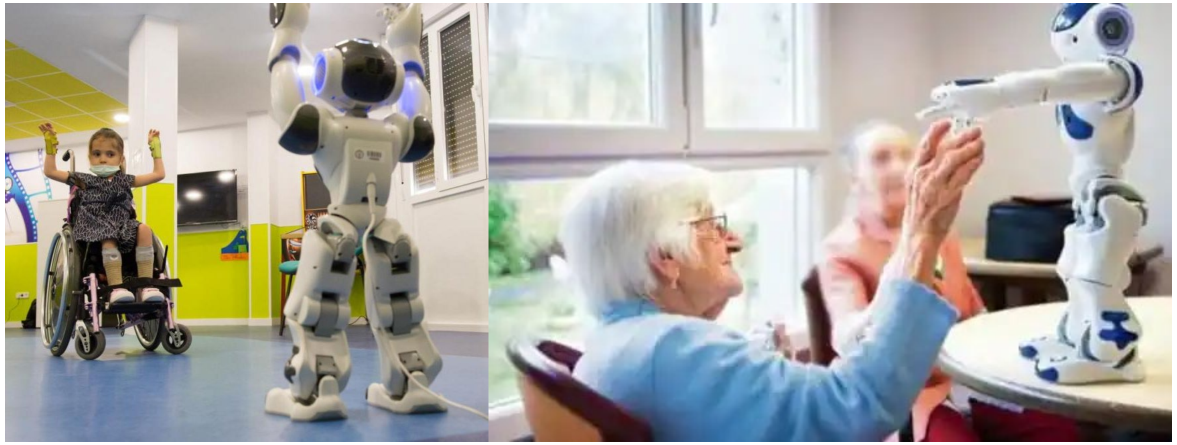
Medical imaging, healthcare, social robots



Skin cancer classification with deep learning
https://cs.stanford.edu/people/esteva/nature/



condition screening, rehabilitation

assisted living, patient monitoring [Lan et al, PAMI 2012]

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Virtual Reality (VR) & Augmented Reality (AR)

INVESTING 3/25/2014 @ 5:43PM | 70,399 views

Facebook Buys Oculus, Virtual Reality Gaming Startup, For $2 Billion

+ Comment Now + Follow Comments

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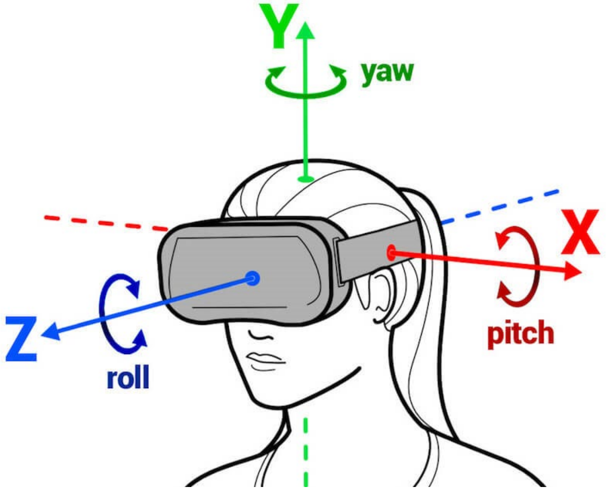
Virtual Reality (VR) & Augmented Reality (AR)



6DoF head tracking

Hand & body tracking

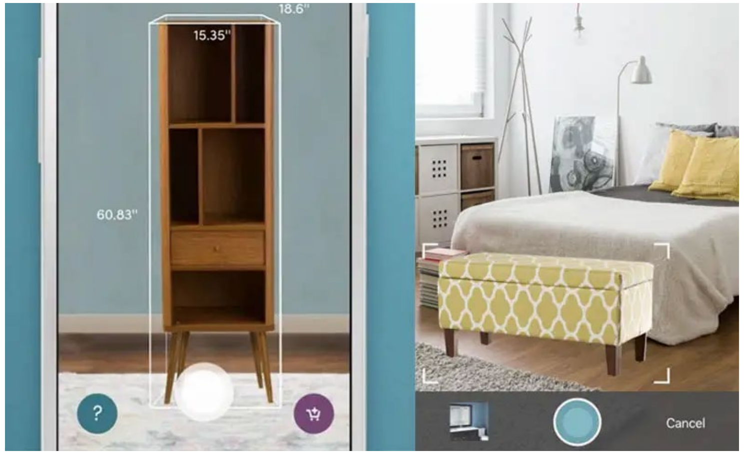
3D scene understanding

3D-360 video capture

VR & AR applications

Microsoft's XBox Kinect

Navigation



Try furniture in your room

Immerse in the room

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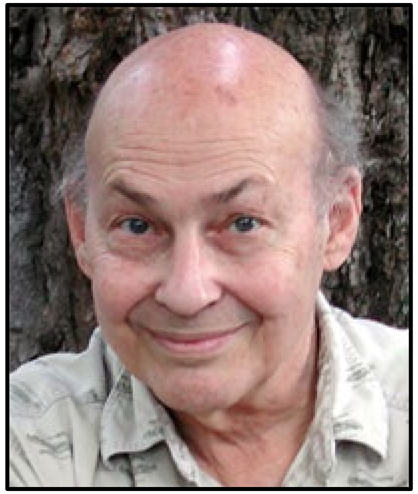
Current state of the art

- Many of these examples are less than 5 years old
- Computer vision is an active research area, and rapidly changing
  - Many new apps in the next 5 years
  - Deep learning powering many modern applications
- Many startups across a dizzying array of areas
  - Deep learning, robotics, autonomous vehicles, medical imaging, construction, inspection, VR/AR, …
- A website of computer vision industries maintained by Prof. David Lowe (UBC): http://www.cs.ubc.ca/~lowe/vision.html

How hard is computer vision?



Marvin Minsky, MIT

Turing award,1969

"In 1966, Minsky hired a first-year undergraduate student and assigned him a problem to solve over the summer: connect a television camera to a computer and get the machine to describe what it sees."

Crevier 1993, pg. 88

MASSACHUSETTS INSTITUTE OF TECHNOLOGY

PROJECT MAC

Artificial Intelligence Group

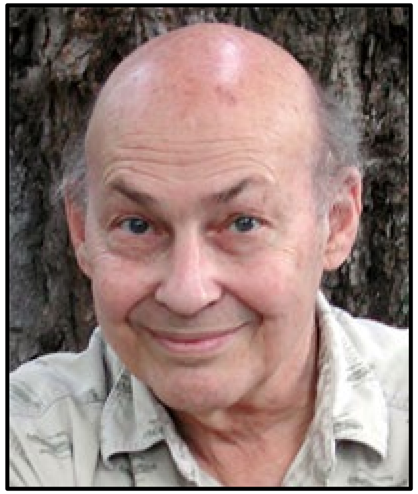
July 7, 1966

Vision Memo. No. 100.

THE SUMMER VISION PROJECT

Seymour Papert

The summer vision project is an attempt to use our summer workers effectively in the construction of a significant part of a visual system. The particular task was chosen partly because it can be segmented into sub-problems which will allow individuals to work independently and yet participate in the construction of a system complex enough to be a real landmark in the development of "pattern recognition".



Marvin Minsky, MIT

Turing award,1969

Gerald Sussman, MIT

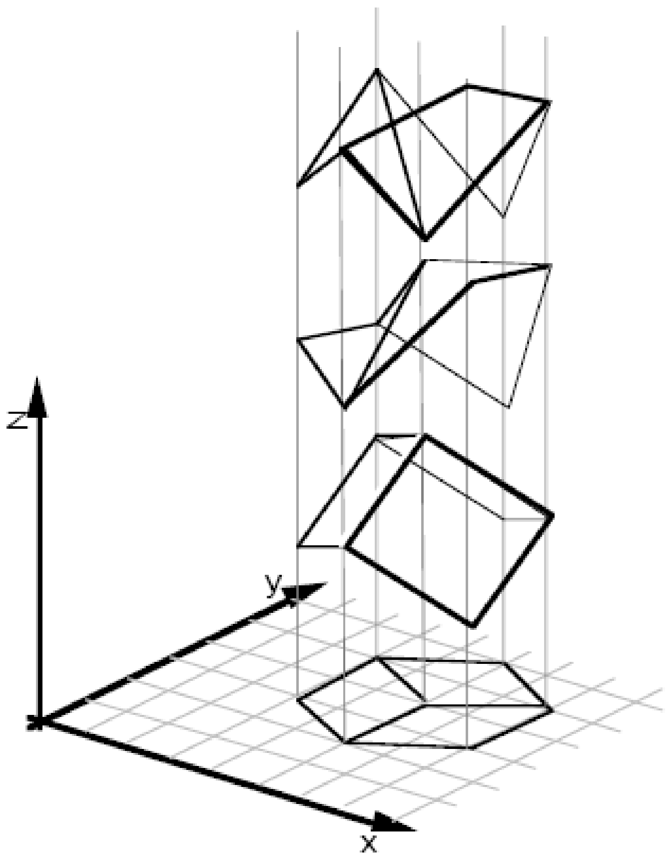
(the undergraduate)

"You'll notice that Sussman never worked in vision again!" – Berthold Horn

Why is computer vision hard?

Why is computer vision difficult?

- Ill-posed problem



[Sinha and Adelson 1993]

Why is computer vision difficult?

Viewpoint variation

Illumination

Scale

Why is computer vision difficult?

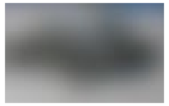
Intra-class variation

Motion (Source: S. Lazebnik)

Background clutter

Occlusion

Challenges: local ambiguity



slide credit: Fei-Fei, Fergus & Torralba

Bottom line

- Perception is an inherently ambiguous problem
  - Many different 3D scenes could have given rise to a given 2D image

Artist Julian Beever with his anamorphic Coke bottle

  - We often must use prior knowledge about the world's structure
  - And apply constraints within your application…

Key considerations:

Data containing the **relevant patterns** is needed!

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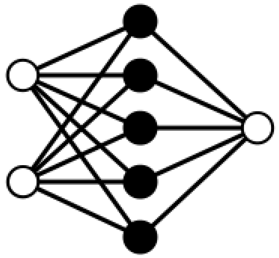
Key considerations:

Data Quality: Poor or biased data leads to inaccurate predictions

Good data

TRAIN



DEPLOY

TEST

Not so good data

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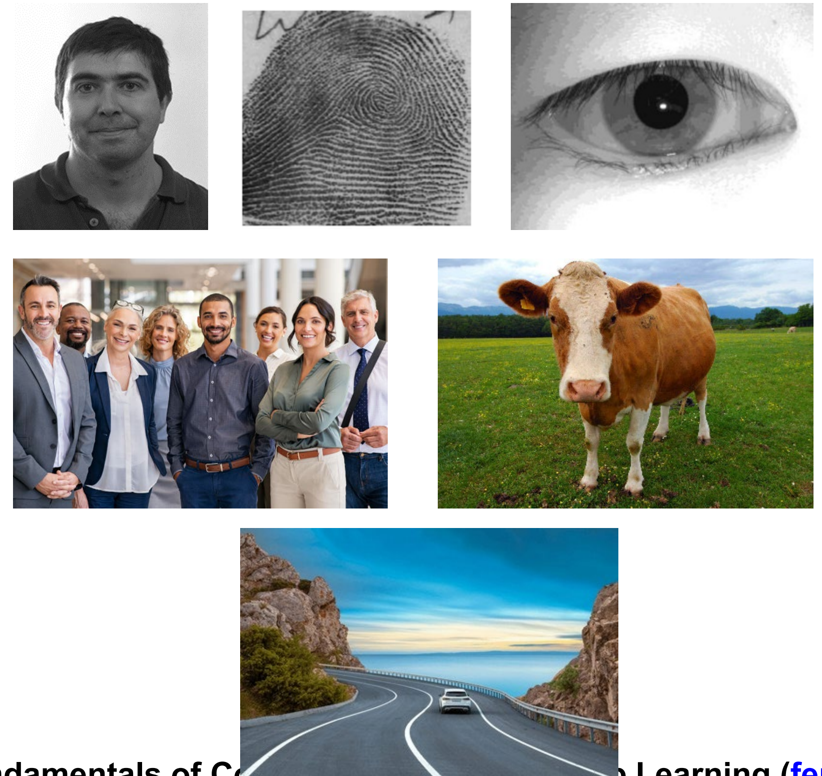
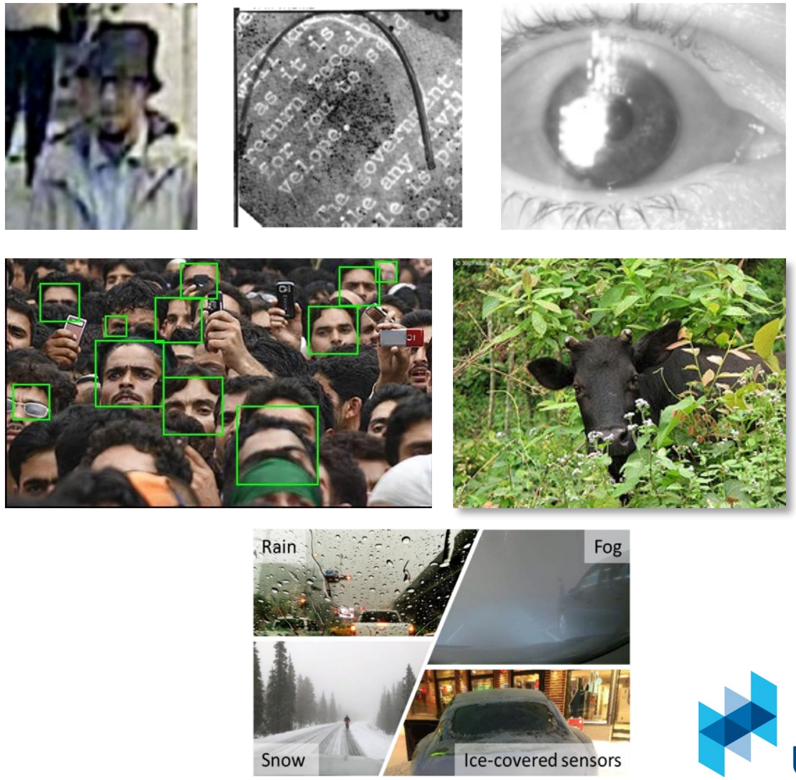
Key considerations:

Data Quality: Poor or biased data leads to inaccurate predictions

Good data

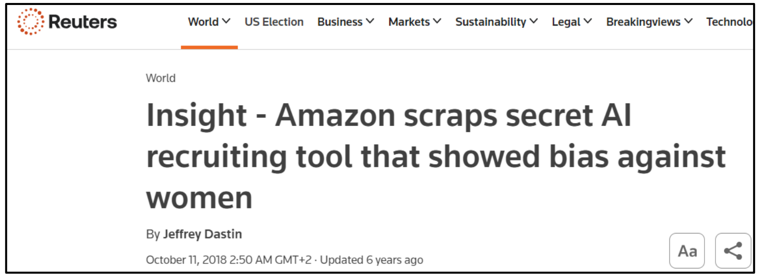


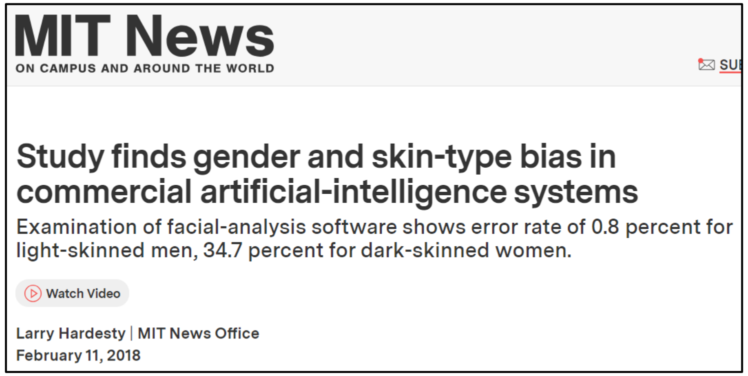
Not so good data



Key considerations:

Data Quality: Poor or **biased** data leads to inaccurate predictions





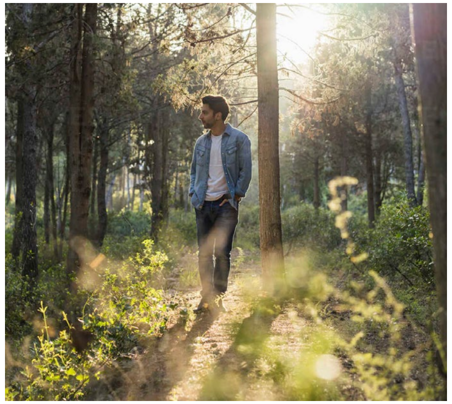
Biases an inequalities in society can (inadvertently) be passed on to AI systems

https://sitn.hms.harvard.edu/flash/2020/racial-discrimination-in-face-recognition-technology/

Key considerations:

Interpretability: Decisions of ML models can be hard to explain

?

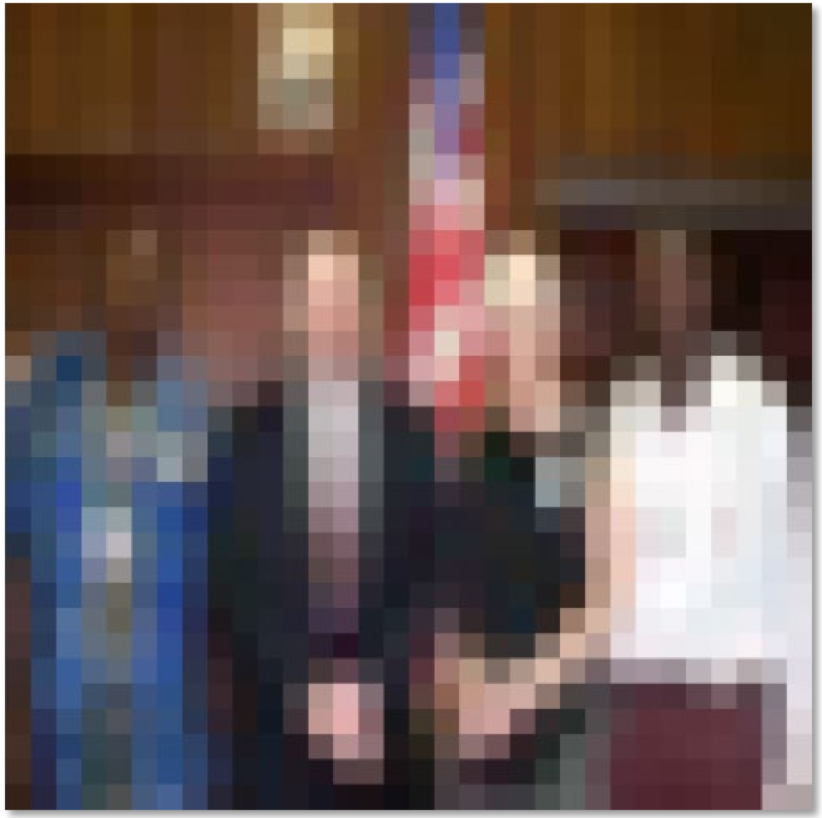


human

moose

?

Humans can tell a lot about from a little information…



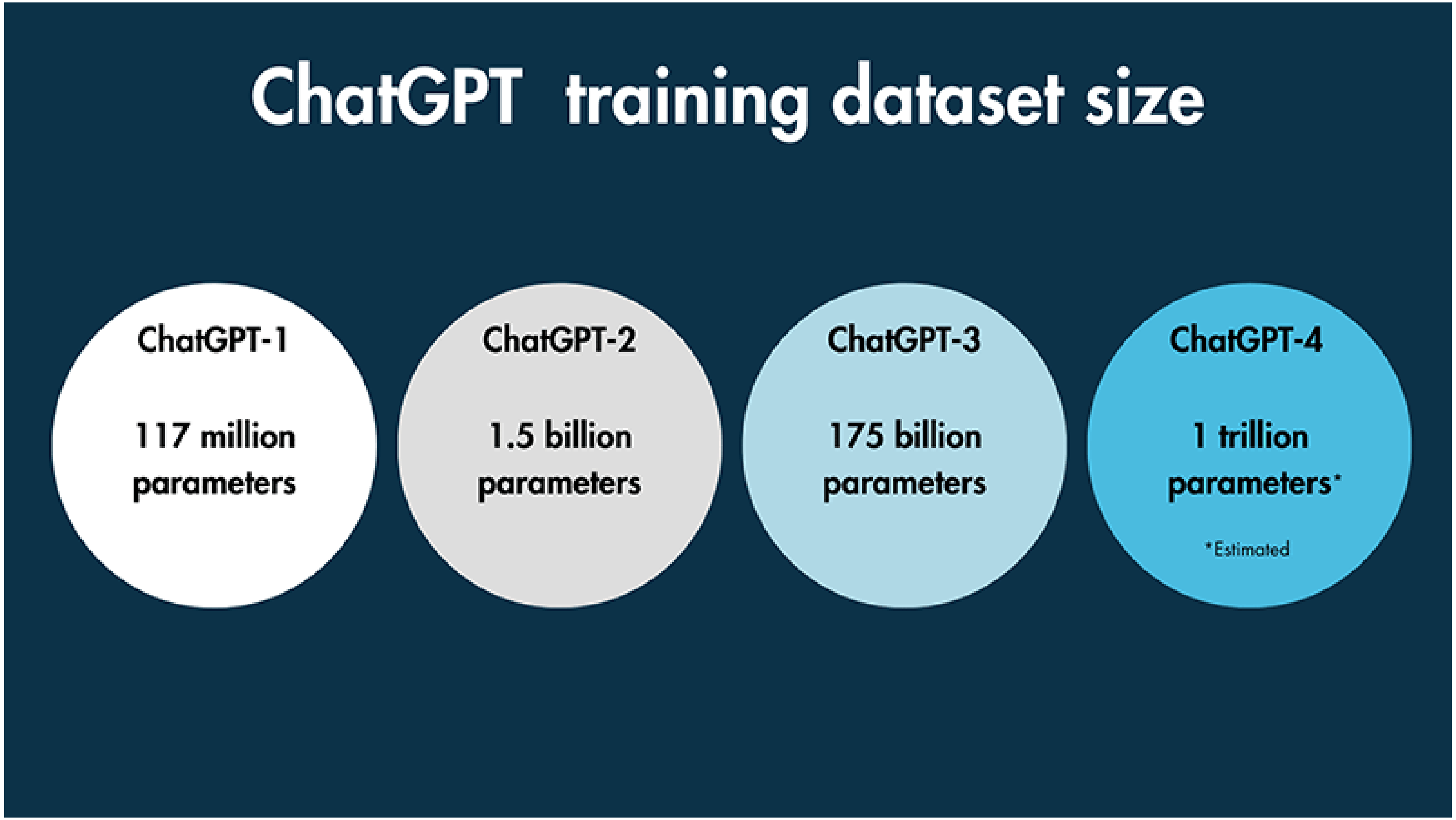
Source: "80 million tiny images" by Torralba, et al.

The state of Computer Vision and AI: we are really, really far.

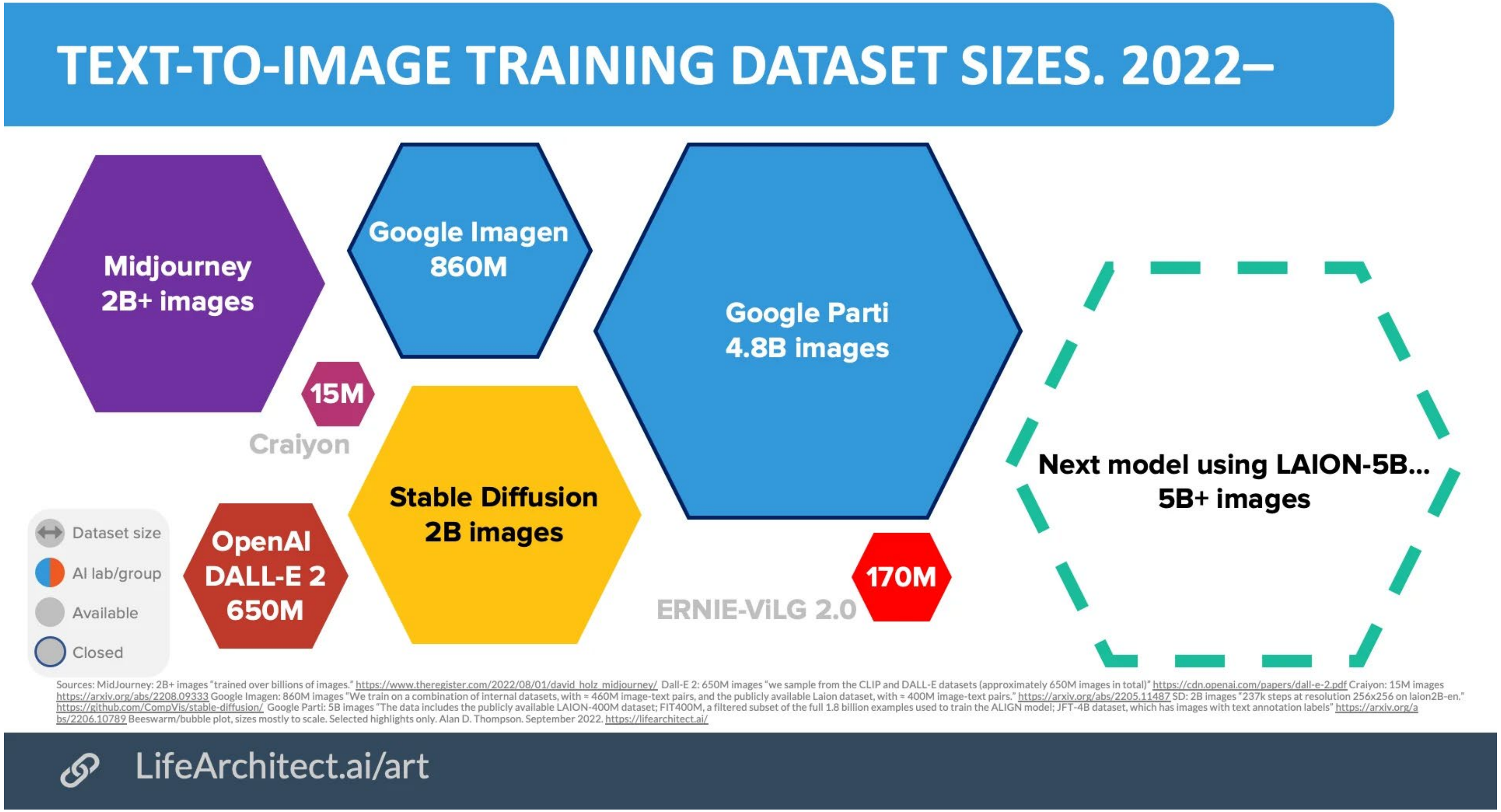
Oct 22, 2012

The picture above is funny.

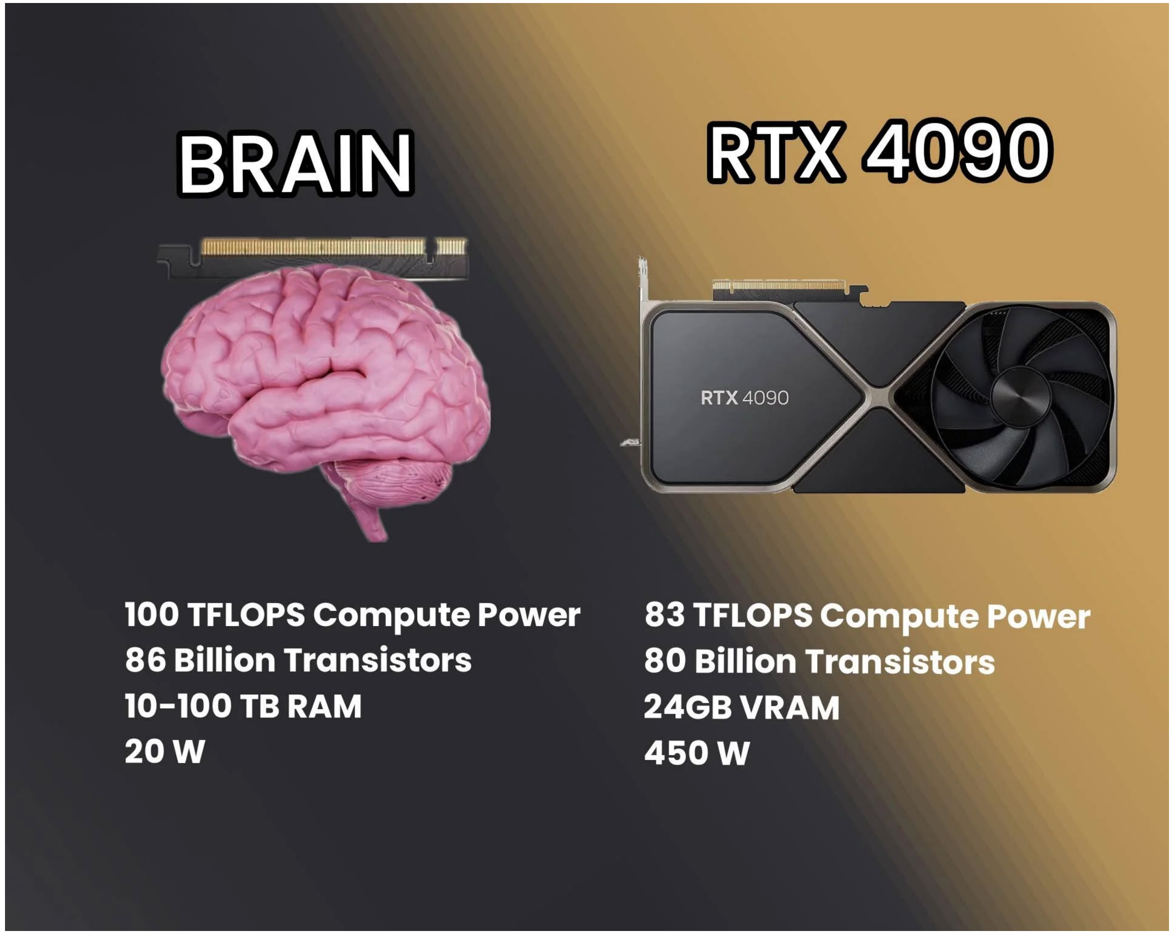
Machines typically cannot…



Machines typically cannot…



Machines typically cannot...



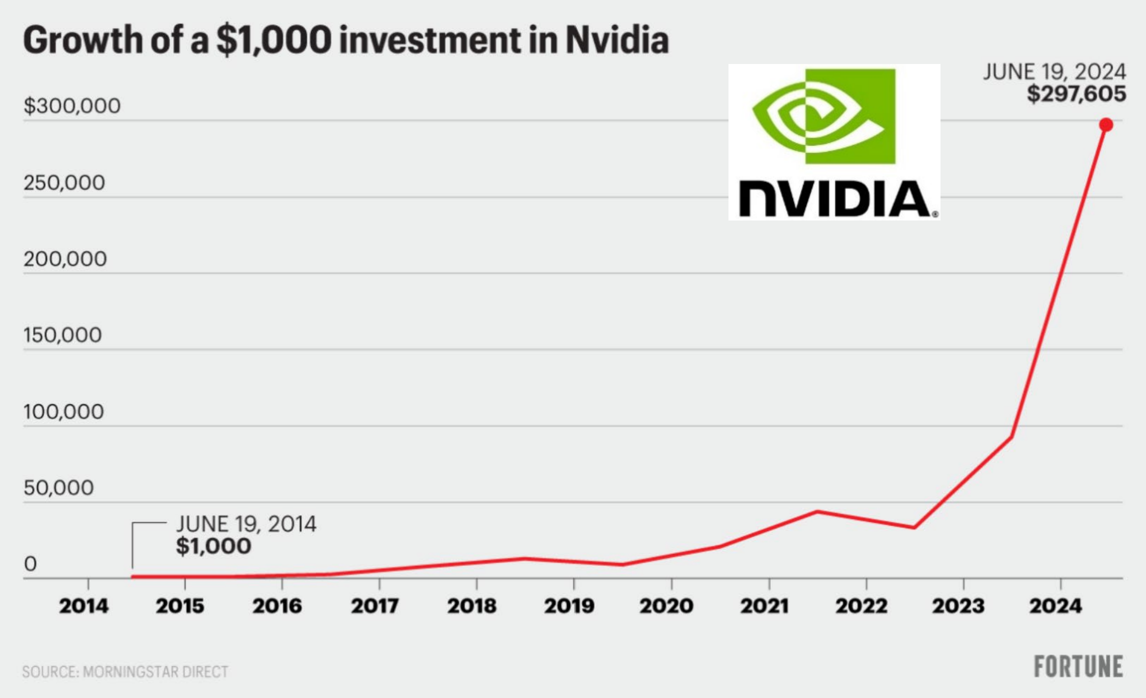
- Some ChatGPT commentators have estimated that if ChatGPT was to be trained on a single NVIDIA Tesla V100 'Graphics Processing Unit' (GPU) that it would take around 355 years to complete ChatGPT's training on its training dataset. However, OpenAI reportedly used 1,023 A100 GPUs to train ChatGPT, so **it is possible that the training process was completed in as little as 34 days**. (Source: Lambda Labs.)

*** A100 GPU is 4x more powerful than RTX 4090**

RTX 4090: 1.6k-2k USD

A100: 10k-15k USD

An AI Hype?



GP-WHO?

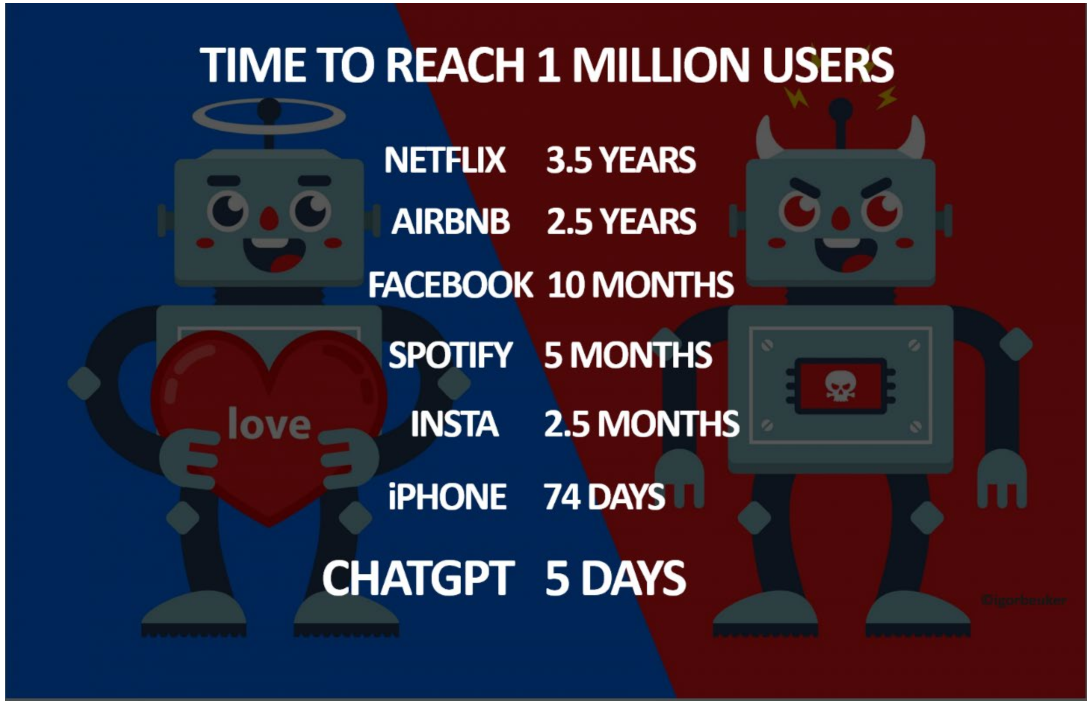
Just four companies are hoarding tens of billions of dollars worth of Nvidia GPU chips

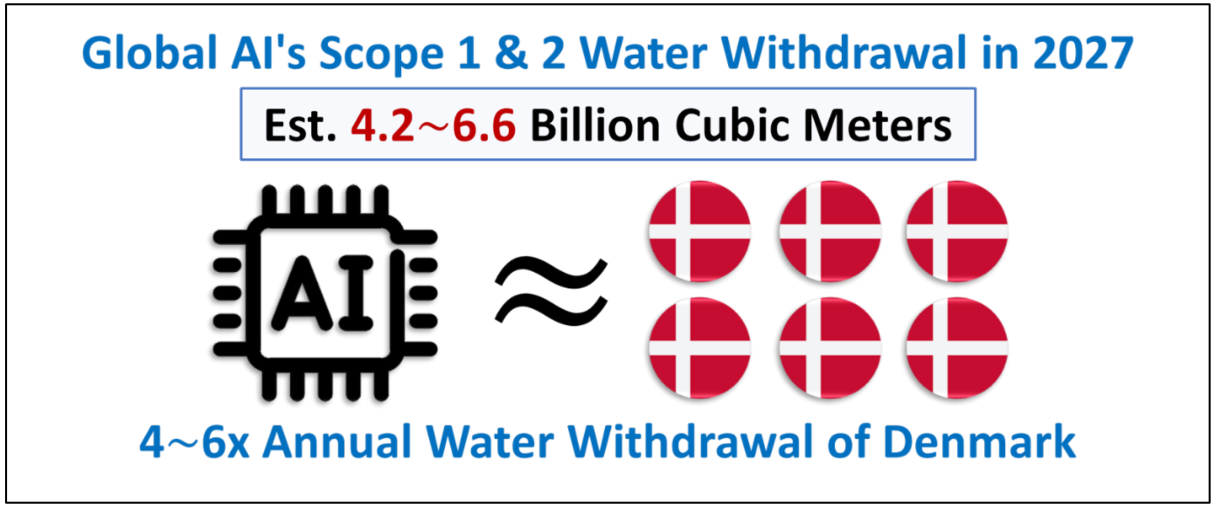
Each Nvidia H100 can cost up to $40,000, and one big tech company has 350,000 of them.

Jon Keegan 7/25/24 1:26PM

- Availability of powerful GPUs (Graphical Processing Units) and cheap storage

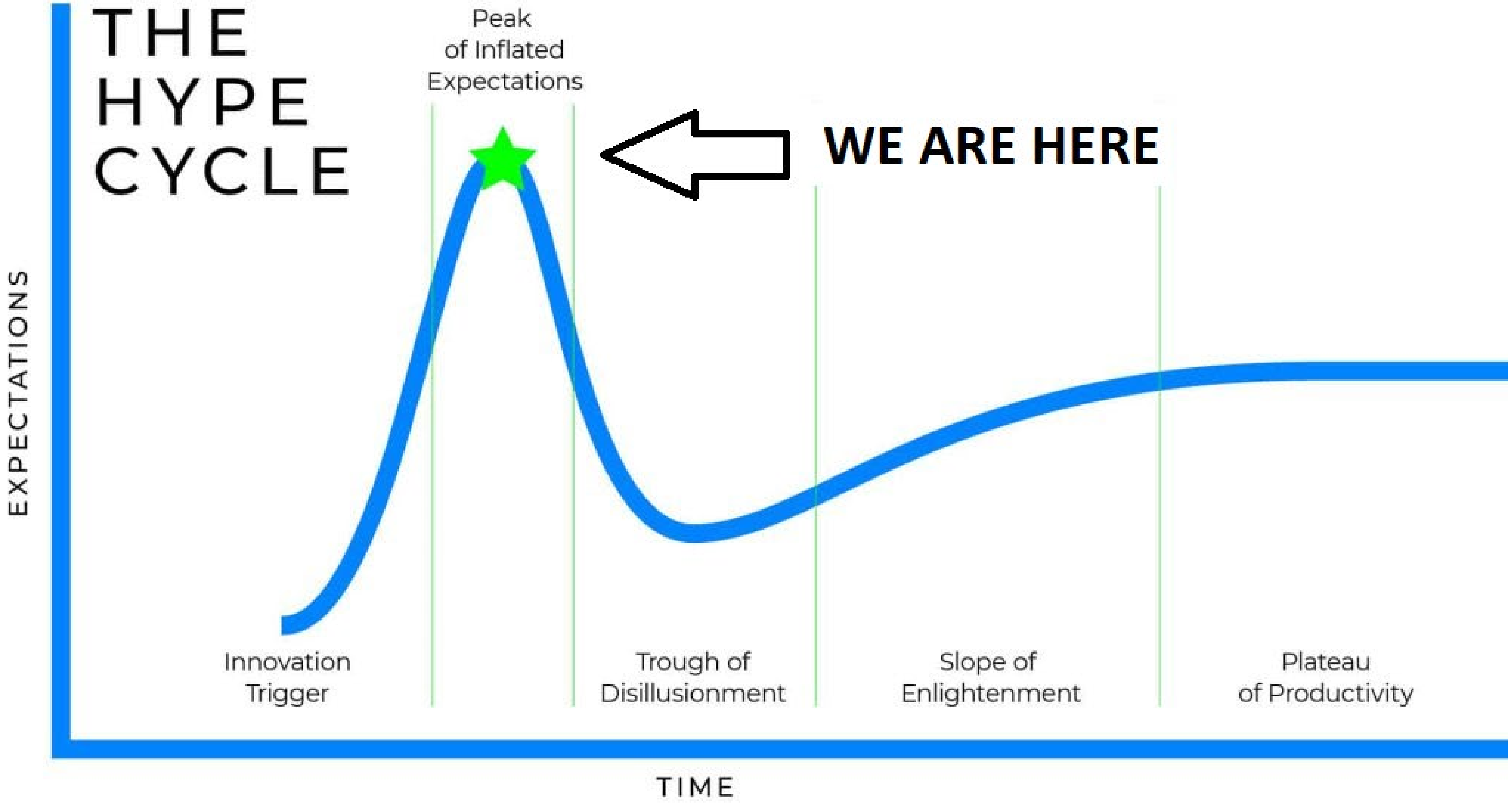
An AI Hype?



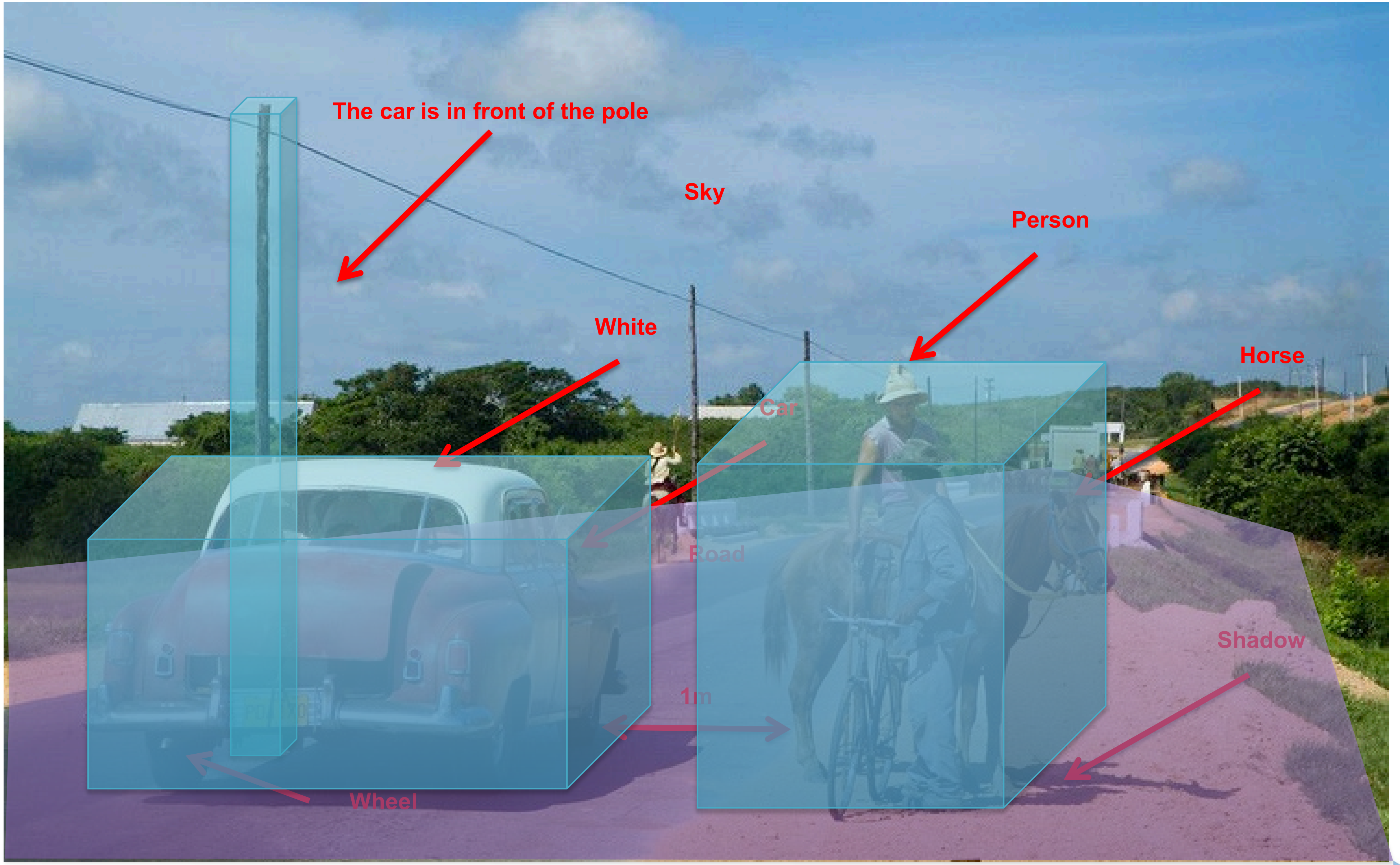


- Availability of powerful GPUs (Graphical Processing Units) and cheap storage

An AI Hype?


THE
HYPE
CYCLE
Peak
of Inflated
Expectations
WE ARE HERE
EXPECTATIONS
Innovation
Trigger
Trough of
Disillusionment
Slope of
Enlightenment
Plateau
of Productivity
TIME

Computer Vision levels


The car is in front of the pole
Sky
Person
White
Horse
Car
Road
Shadow
1m
Wheel

Computer Vision

- Low Level Vision
  - Measurements
  - Preprocessing
  - Enhancements
- Mid Level Vision
  - Segmentation
  - Reconstruction
  - Depth, motion
  - Feature extraction
- High Level Vision
  - Category detection
  - Activity recognition
  - Deep understandings

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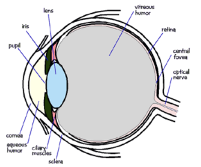
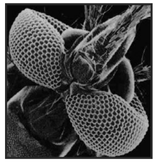
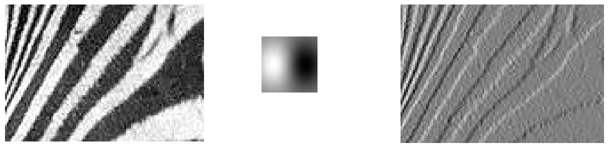



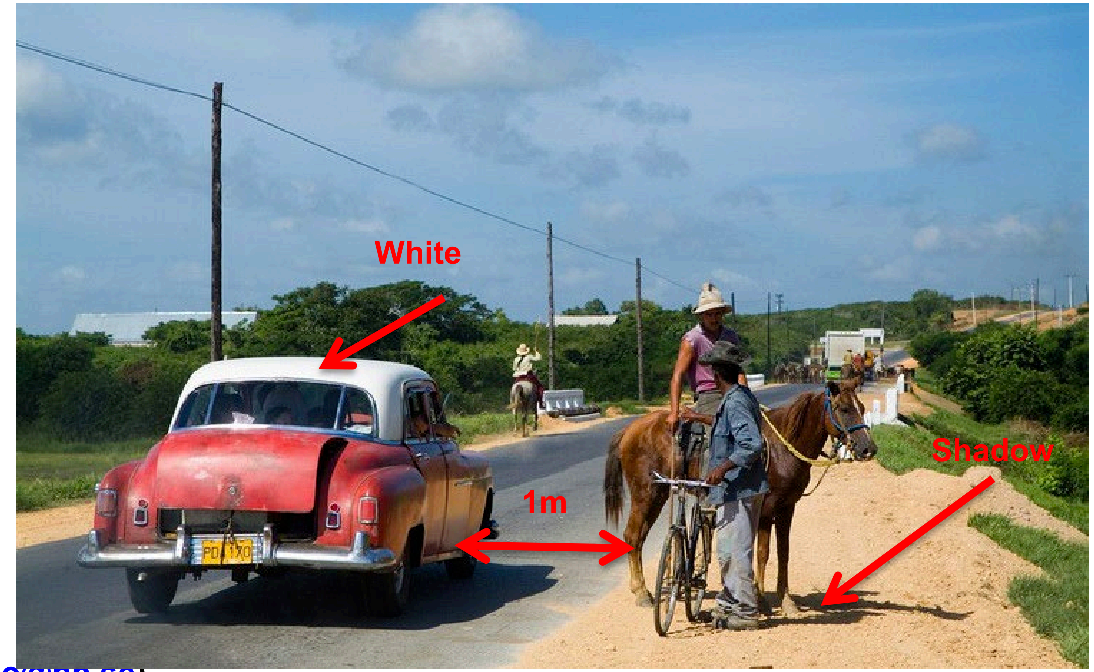
Image formation



Filtering, edge detection
Sharpening

Noise reduction
Restoration
Enhancement

Morphological operations



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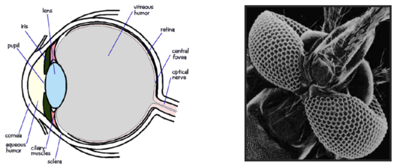
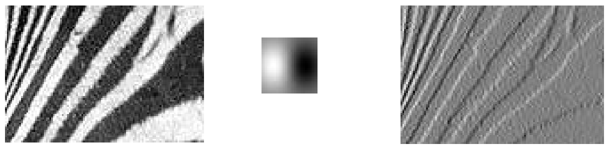


Image formation


Filtering, edge detection
Sharpening


Noise reduction
Restoration
Enhancement

Morphological operations

Low-Level processing

– *Primitive* initial operations

– Result = another image with the operation applied

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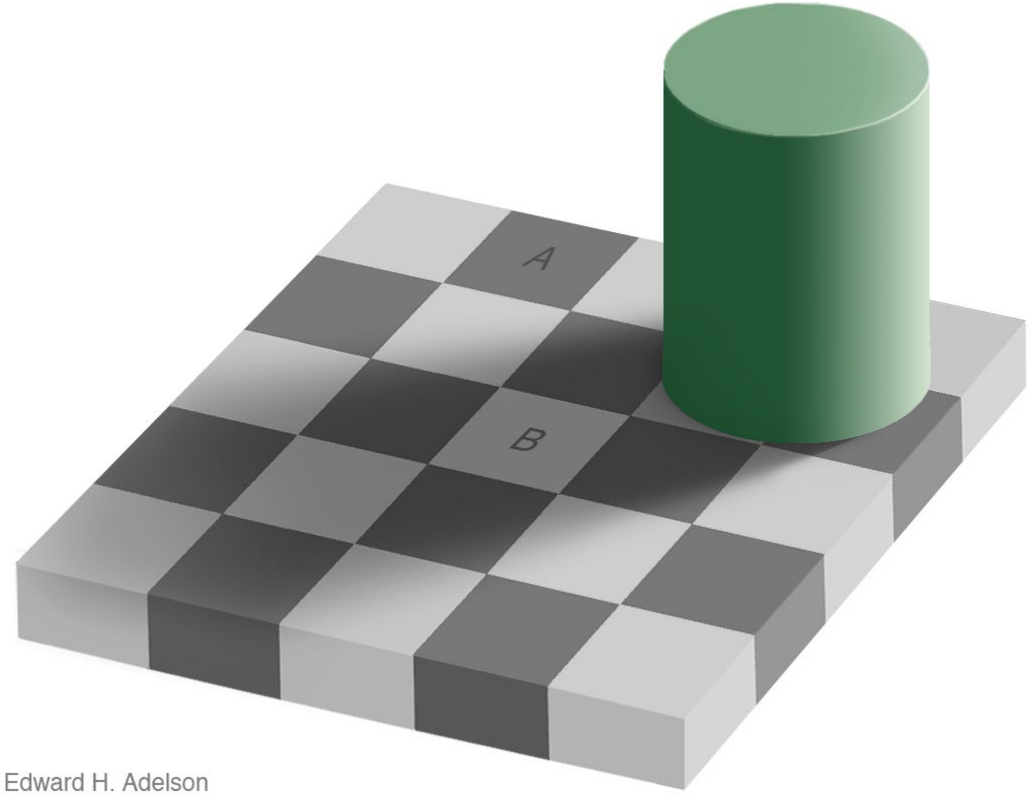
Measurement can be problematic…

Brightness and shadows



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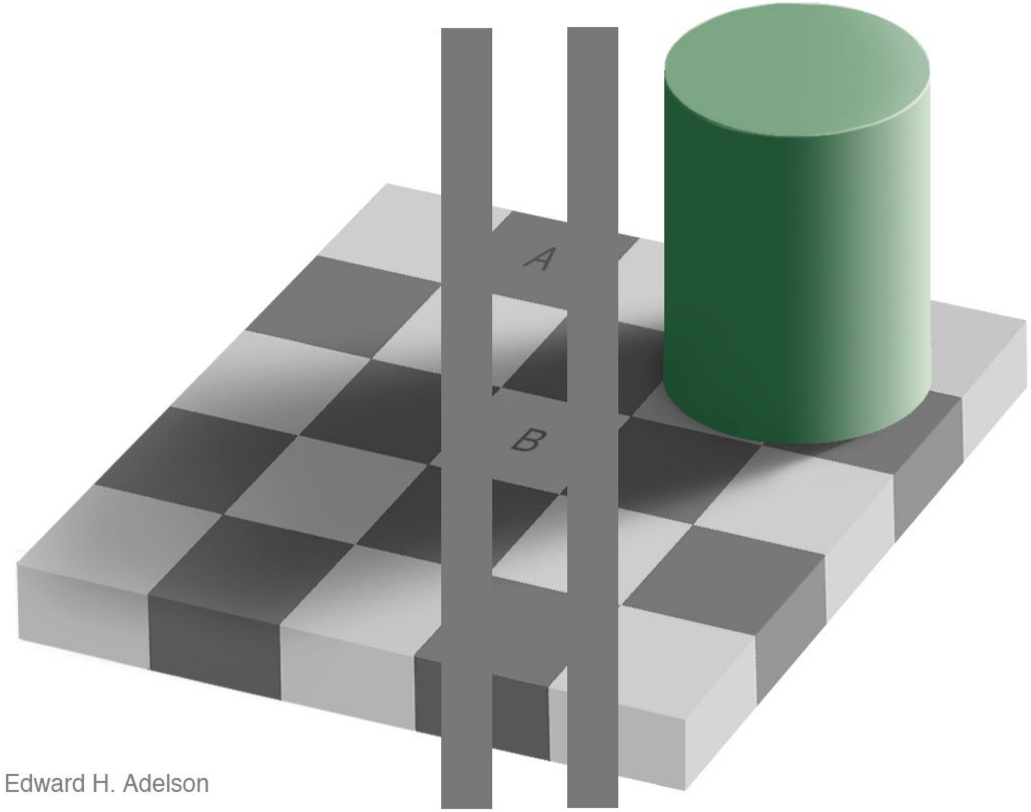
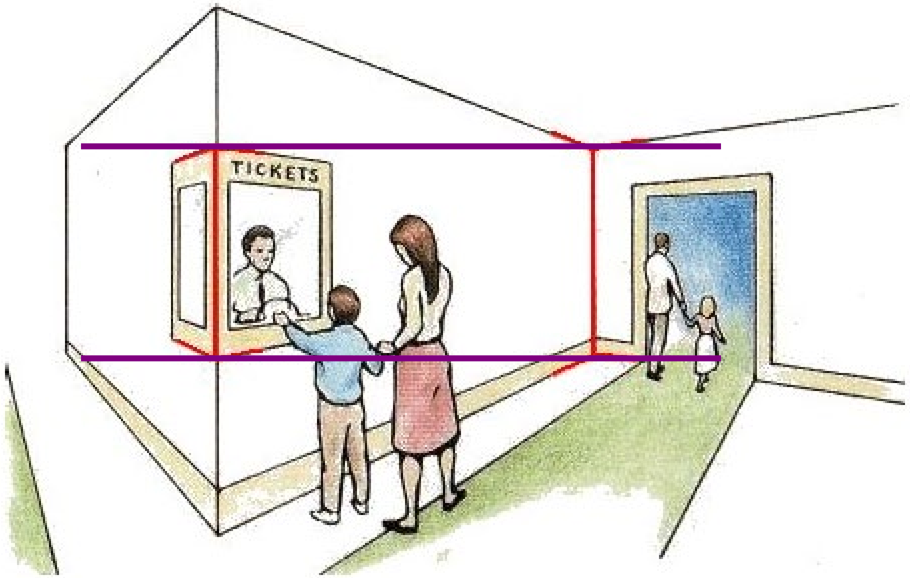
Brightness and shadows



http://www.newworldencyclopedia.org/entry/Same_color_illusion

Measurement can be problematic...

Length



Müller-Lyer Illusion

http://www.michaelbach.de/ot/sze_muelue/index.html

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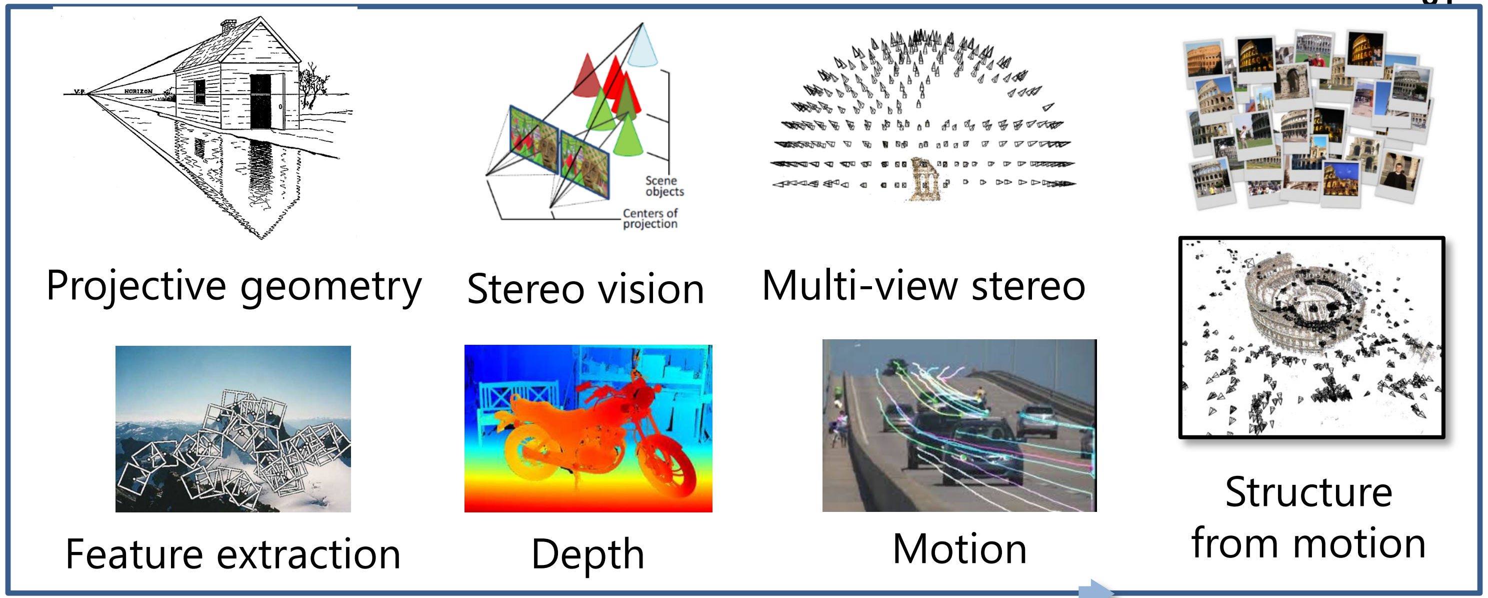

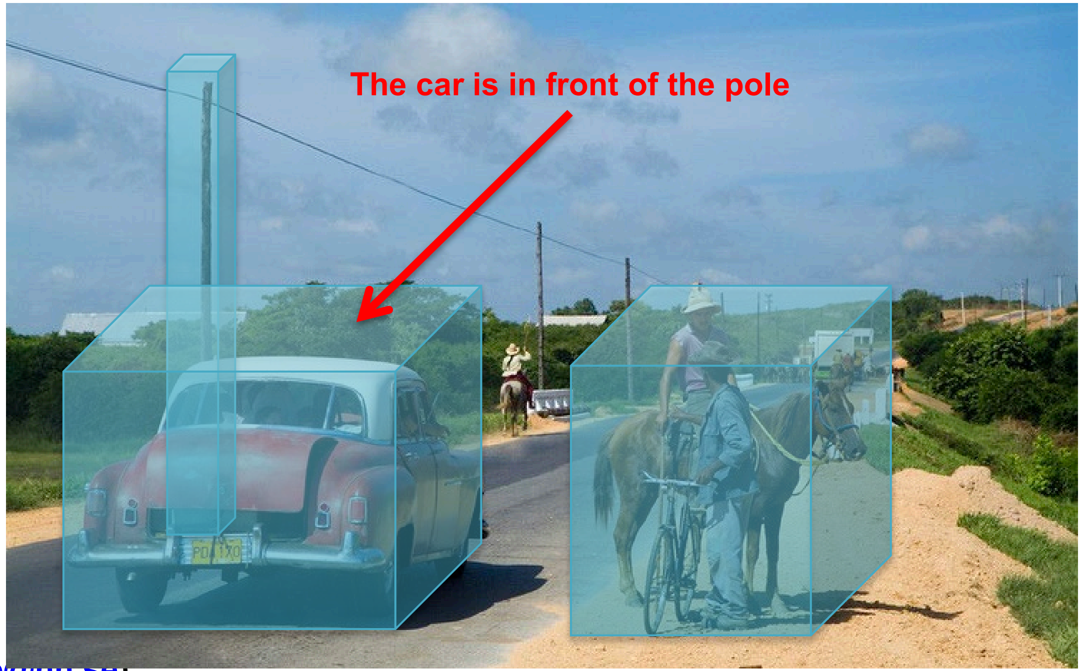

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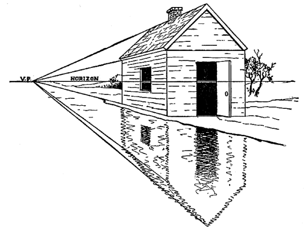
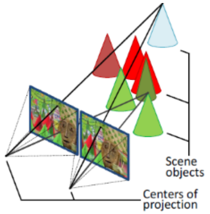
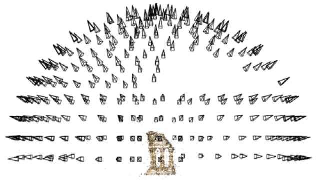


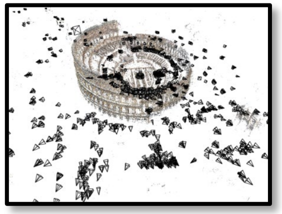
Projective geometry



Stereo vision



Multi-view stereo



Structure from motion

Feature extraction

Depth

Motion

Mid-Level processing

– Result = features or attributes describing the image (regions, objects, motion…)

Measurement can be problematic…

Perceived motion vs. real motion

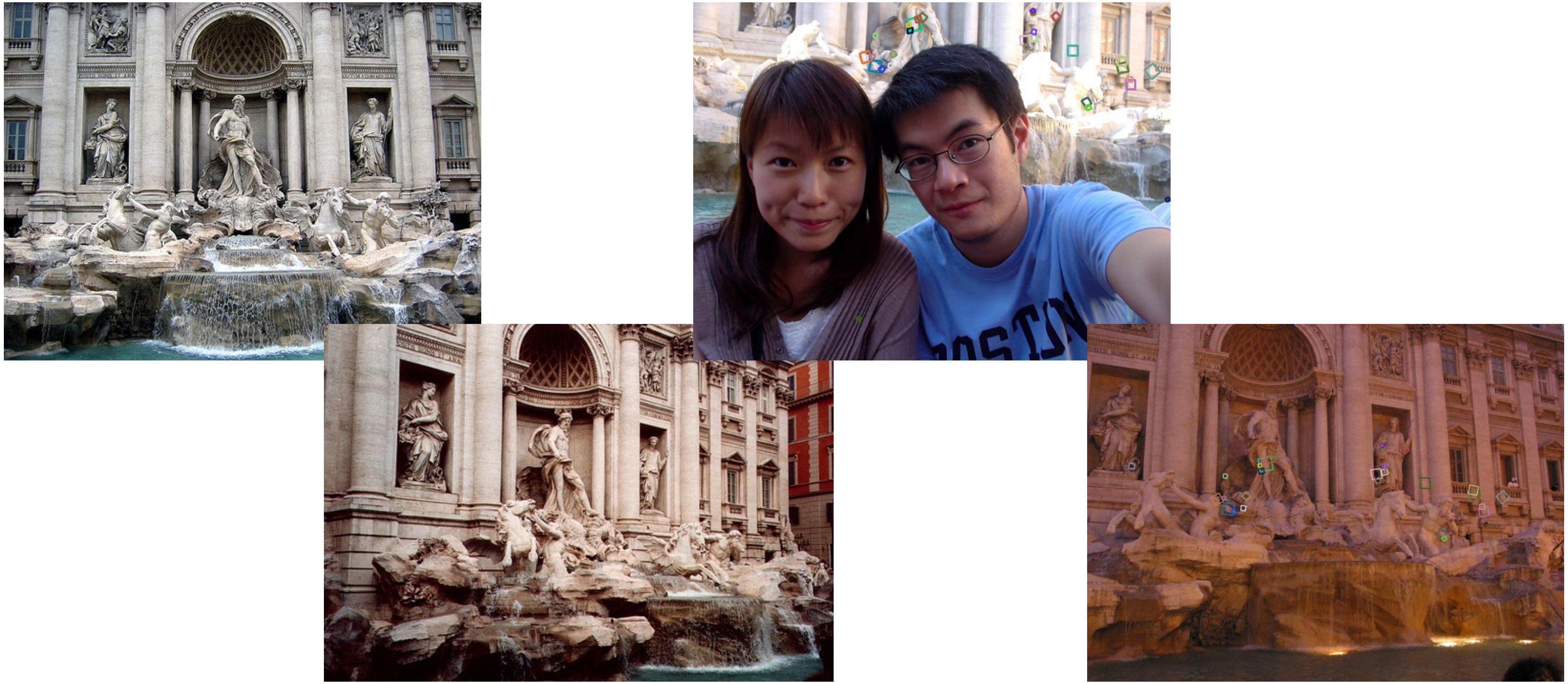
Measurement can be problematic…

Perceived motion vs. real motion

84

Measurement can be problematic…

Impact of covariates



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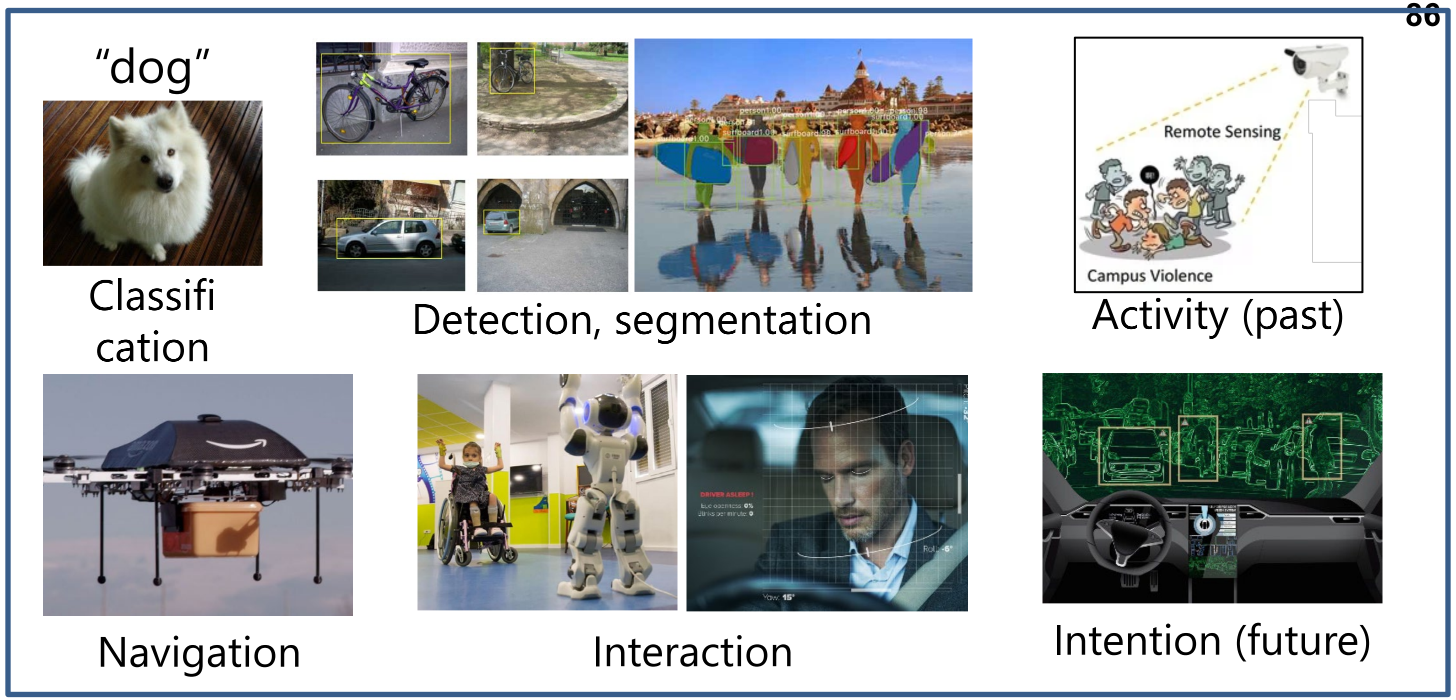

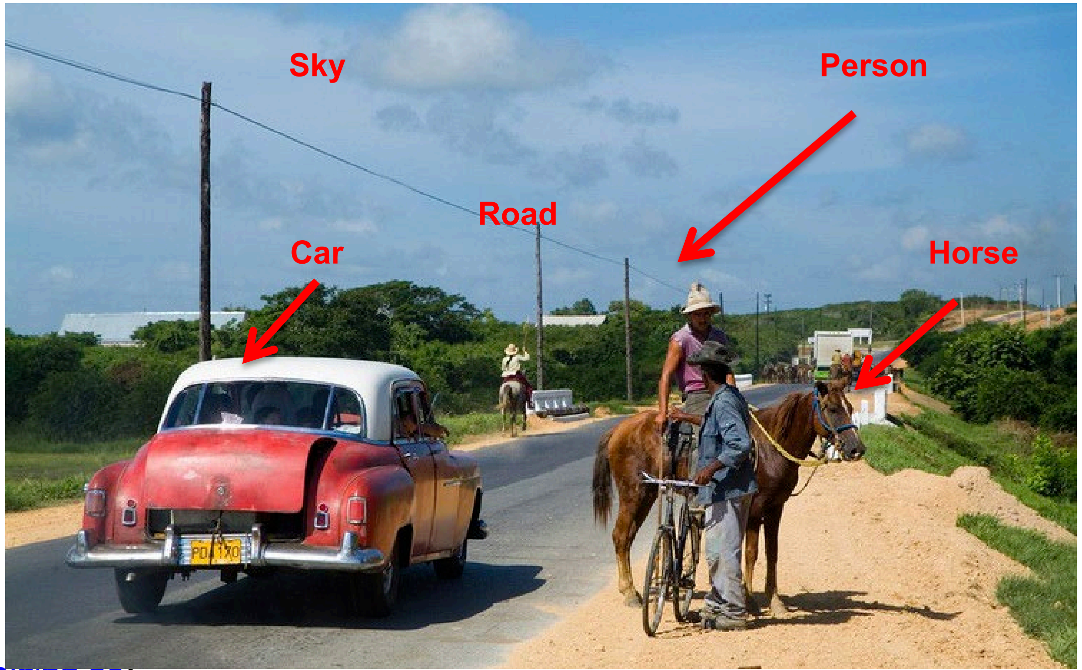

Computer Vision

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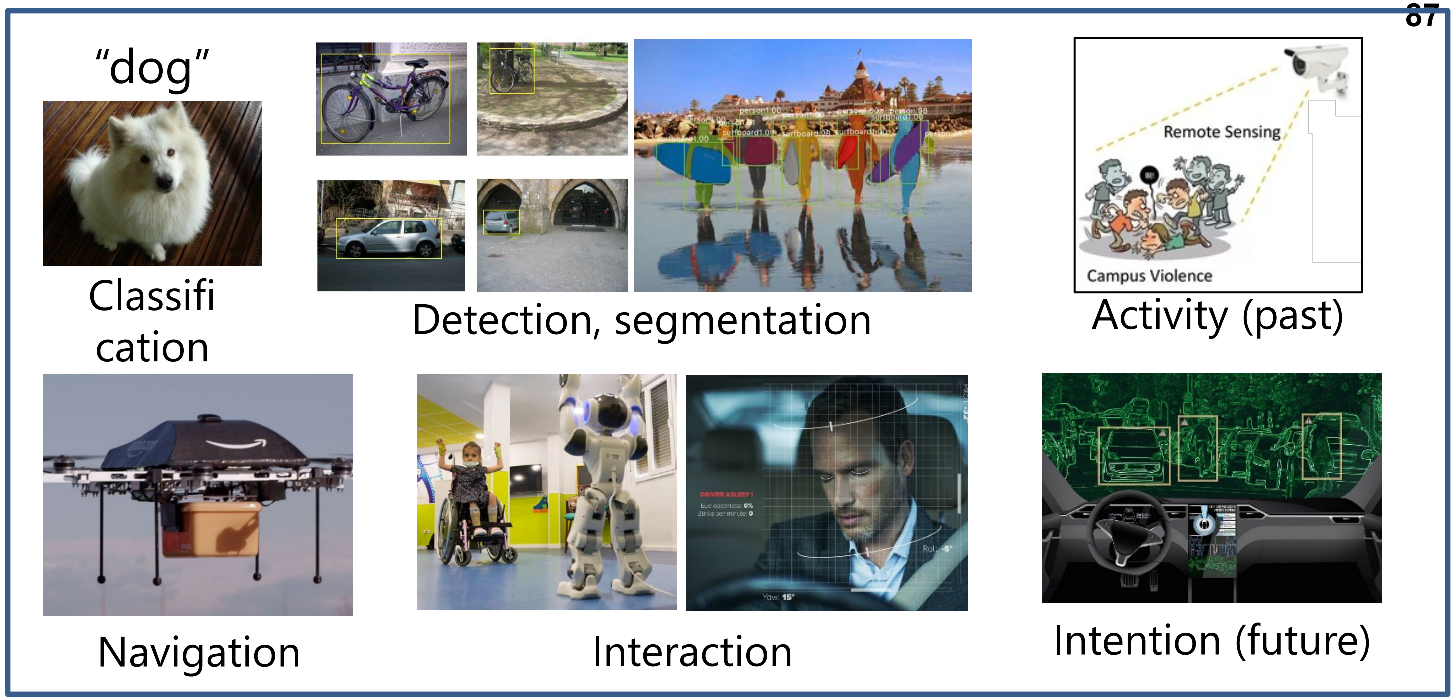

High-Level processing

- Result = "making sense" of the image, "cognitive" vision
- Totally dependent of the task

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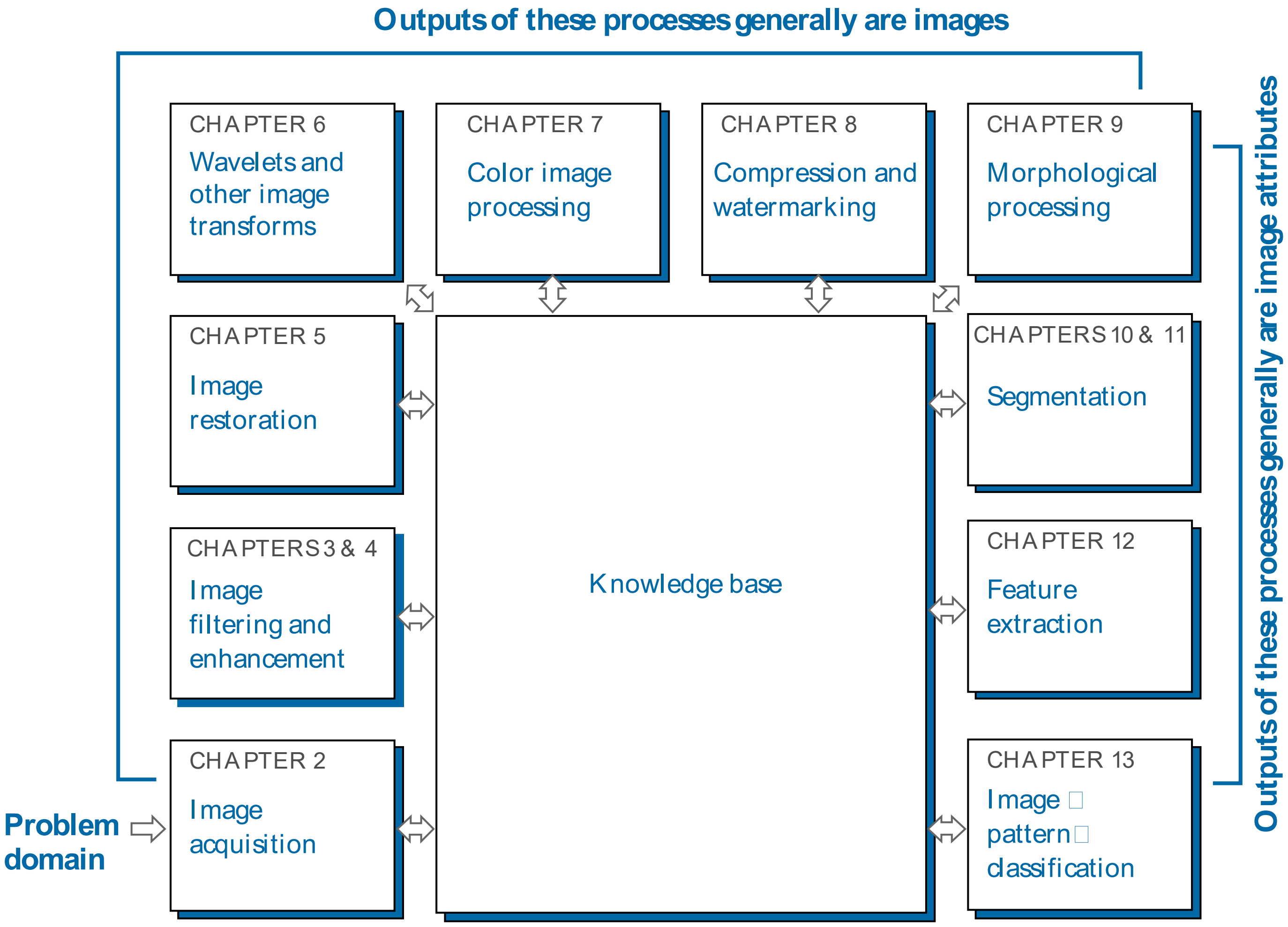
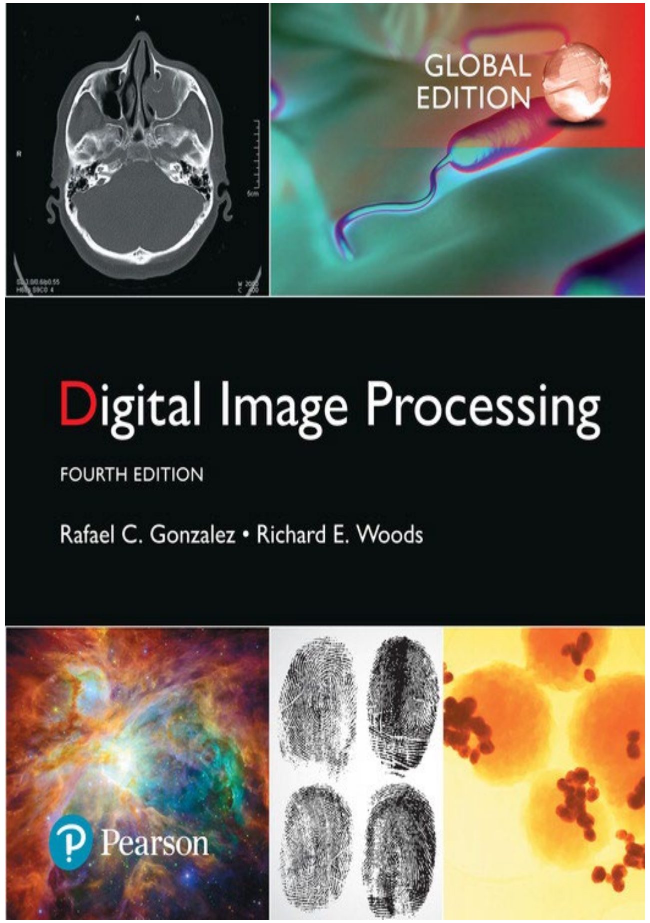
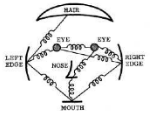
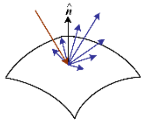
Gonzalez book

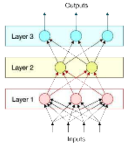
FIGURE 1.23 Fundamental steps in digital image processing. The chapter(s) indicated in the boxes is where the material described in the box is discussed.



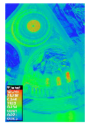
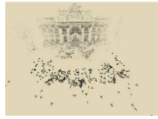
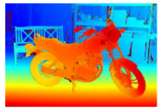


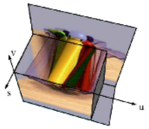
Szeliski book

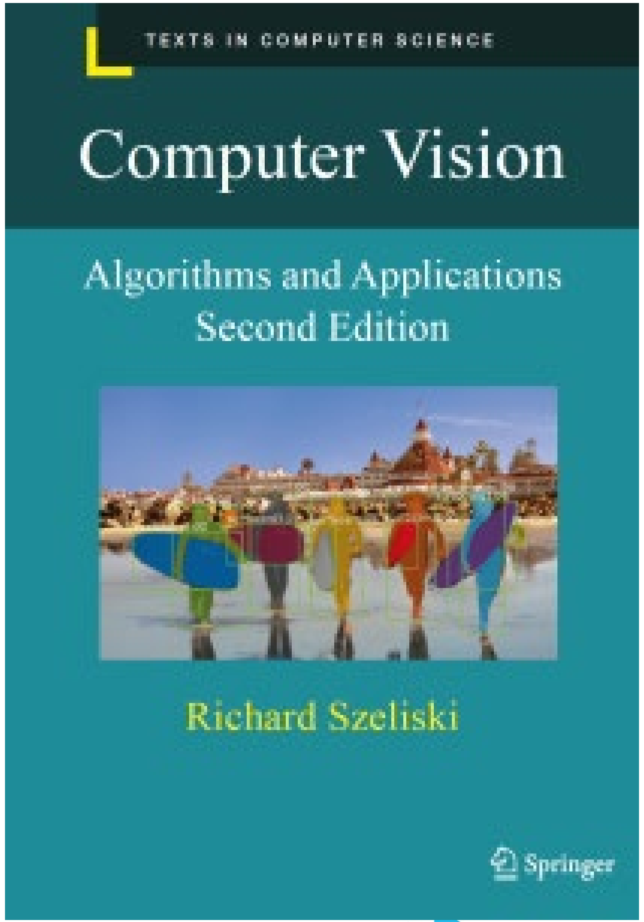



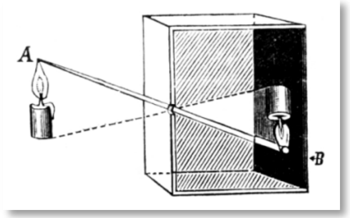


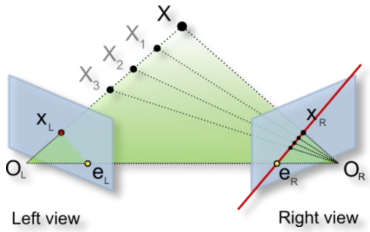




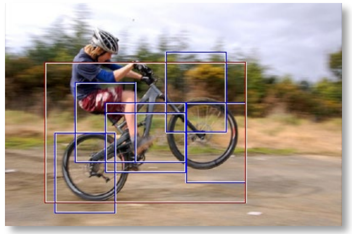
Course Overview



1. Low-level vision: *primitive* operations
   – Image acquisition and representation, transformations, enhancements, convolution and filtering, edges and lines…



2. Mid-level vision: *attribute* extraction
   ~~– Segmentation, depth, motion, 3D…~~ (not covered)
   – Feature extraction, classification, image matching…
   – Machine Learning (ML), Convolutional Neural Nets (CNN)



3. High-level vision: *making sense* of the image
   – Object / category / activity… detection and recognition

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Questions?